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Concurrent exercise incorporating high-intensity interval or continuous training modulates mTORC1 signaling and microRNA expression in human skeletal muscle

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Fyfe JJ, Bishop DJ, Zacharewicz E, Russell AP, Stepto NK. Concurrent exercise incorporating high-intensity interval or continuous training modulates mTORC1 signaling and microRNA expression in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 310: R1297–R1311, 2016. First published April 13, 2016; doi:10.1152/ajpregu.00479.2015.—We compared the effects of concurrent exercise, incorporating either high-intensity interval training (HIT) or moderate-intensity continuous training (MICT), on mechanistic target of rapamycin complex 1 (mTORC1) signaling and microRNA expression in skeletal muscle, relative to resistance exercise (RE) alone. Eight males (mean $\pm$ SD: age, 27 ± 4 yr; $\dot{V}O_{2\text{ peak}}$, 45.7 ± 9 ml·kg⁻¹·min⁻¹) performed three experimental trials in a randomized order: 1) RE (8 $\times$ 5 leg press repetitions at 80% 1-repetition maximum) performed alone and RE preceded by either 2) HIT cycling [10 $\times$ 2 min at 120% lactate threshold (LT); HIT + RE] or 3) work-matched MICT cycling (30 min at 80% LT; MICT + RE). Vastus lateralis muscle biopsies were obtained immediately before RE, either without (REST) or with (POST) preceding endurance exercise and +1 h (RE + 1 h) and +3 h (RE + 3 h) after RE. Prior HIT and MICT similarly reduced muscle glycogen content and increased ACC^{Ser79} and p70S6K^{Thr389} phosphorylation before subsequent RE (i.e., at POST). Compared with MICT, HIT induced greater mTOR^{Ser2448} and rps6^{Ser235/236} phosphorylation at POST. RE-induced increases in p70S6K and rps6 phosphorylation were not influenced by prior HIT or MICT; however, mTOR phosphorylation was reduced at RE + 1 h for MICT + RE vs. both HIT + RE and RE. Expression of miR-133a, miR-378, and miR-486 was reduced at RE + 1 h for HIT + RE vs. both MICT + RE and RE. Postexercise mTORC1 signaling following RE is therefore not compromised by prior HIT or MICT, and concurrent exercise incorporating HIT, but not MICT, reduces postexercise expression of miRNAs implicated in skeletal muscle adaptation to RE.

concurrent training; interference; exercise intensity; high-intensity interval training; continuous training

INCORPORATING BOTH RESISTANCE (RE) and endurance exercise into a periodized training program is termed concurrent training (56). Compared with undertaking RE alone, concurrent training attenuates skeletal muscle hypertrophy and maximal strength development in some (10, 19, 33, 48, 53), but not all (5, 58, 63, 75, 76), studies. Given that skeletal muscle mass plays an important role in overall metabolic health (87), and many athletes require elements of strength and muscle hypertrophy, concomitantly with a high aerobic capacity (47), min-

imizing interference during concurrent training has implications for optimizing both health and performance outcomes.

The mechanistic target of rapamycin complex 1 (mTORC1) is a key mediator of load-induced increases in muscle protein synthesis (MPS) (13, 32). The activity of mTORC1 is antagonized by activation of the 5'-adenosine monophosphate-activated protein kinase (AMPK), which acts to restore perturbations in cellular energy balance by inhibiting anabolic cellular processes, such as protein synthesis, and stimulating catabolism (51). Given that AMPK, acting primarily through the peroxisome proliferator-activated γ -receptor coactivator 1 α (PGC-1 α), has been associated with many skeletal muscle adaptations to chronic endurance training (67), the interference effect with concurrent training may be partly explained by AMPK-mediated inhibition of the mTORC1 signaling pathway. However, several human studies have shown that concurrent exercise does not compromise either postexercise mTORC1 signaling or rates of MPS (1, 2, 17, 29, 69), or even potentiates these responses (57), relative to RE performed alone. Whether these putative interference mechanisms explain the attenuated skeletal muscle adaptations reported following concurrent training in humans (10, 53, 84) remains unclear.

Additional molecular mechanisms with the potential to regulate skeletal muscle adaptations to exercise have recently emerged, including altered microRNA (miRNA) expression (90). miRNAs are small (~20–30 nucleotides in length), noncoding ribonucleic acid (RNA) species that are highly expressed in skeletal muscle (90). The primary function of miRNAs is to decrease protein levels either by repressing gene translation or promoting the degradation of target mRNAs, of which the latter accounts for the majority of miRNA activity (43). Given their purported role in posttranscriptional regulation, miRNAs have emerged as a potential regulator of exercise-induced adaptations in skeletal muscle (90). The expression of various miRNA species in human skeletal muscle is altered following both single bouts of exercise (31, 72, 91) and after short-term exercise training (26, 72). miRNAs have been implicated in the regulation of the IGF-1/Akt, Fox-O1, and myogenesis pathways, all known to play a role in RE-induced adaptations in skeletal muscle (49), although the direct impact of miRNAs on these pathways in response to RE is yet to be experimentally validated. In addition, whether miRNAs are differentially regulated following concurrent exercise, compared with single-mode RE, is currently unknown.

From a practical standpoint, elucidation of the roles of specific training variables (e.g., training intensity, volume, and modality) is required to inform exercise prescription guidelines

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for minimizing potential interference during concurrent training (38). One important practical consideration is the intensity of endurance training employed in a concurrent training program. Recent evidence suggests that high-intensity interval training (HIT) is more effective in improving $\dot{V}O_{2\max}$ (41, 42, 79, 85) and metabolic health markers (79, 85), compared with moderate-intensity continuous training (MICT), while also being more enjoyable (7). However, most studies investigating the early molecular responses to concurrent exercise in humans have employed MICT (e.g., 30–90 min at 65–70% $\dot{V}O_{2\max}$ or 55–70% $W_{\max}$) (17, 22, 29, 57, 81). Whether the intensity of endurance exercise, when performed concurrently with RE, is important in mediating any potential interference effect on postexercise anabolic responses in skeletal muscle is unknown. There are several potential mechanisms by which higher intensity endurance exercise may exacerbate molecular interference during concurrent training, compared with lower-intensity endurance exercise. For example, endurance exercise activates putative inhibitors of MPS, including AMPK and 4E-BP1 (eIF4E binding protein 1), in an intensity-dependent manner, with higher relative exercise intensities associated with greater AMPK and 4E-BP1 activation (70). Higher intensity endurance exercise also induces greater muscle glycogen depletion (40), which is associated with increased AMPK activity (27). Taken together, these factors suggest that performing higher intensity endurance exercise concurrently with RE may be suboptimal with regards to promoting anabolic responses in skeletal muscle following concurrent exercise. The effect of concurrent exercise incorporating divergent endurance training intensities on early postexercise anabolic responses in skeletal muscle has not been directly compared.

We therefore aimed to compare the effects of a single bout of concurrent exercise, incorporating either HIT or work-matched MICT cycling, on mTORC1 signaling and miRNA expression in human skeletal muscle, compared with RE performed alone. Between-trial comparisons were made to determine 1) the effects of prior endurance exercise on these responses before commencing subsequent RE; 2) whether prior endurance exercise altered mTORC1 signaling and miRNA expression after subsequent RE, compared with RE performed alone; and 3) whether these responses were different when RE was performed after HIT compared with MICT. We hypothesized that, compared with MICT, HIT would further 1) increase glycogen depletion and AMPK activity [indexed by AMPK α -subunit and AMPK substrate acetyl-CoA carboxylase (ACC) phosphorylation] before subsequent RE, 2) attenuate mTORC1 signaling after subsequent RE, and 3) alter the postexercise expression of miRNAs linked to the regulation of skeletal muscle adaptations to RE.

METHODOLOGY

Participants

Eleven males participating in regular physical activity incorporating both resistance and endurance exercise (>30 min, 3–5 times per wk) were initially recruited for this investigation; however, three participants withdrew due to circumstances unrelated to the study. Eight participants [mean $\pm$ SD: age, 27 $\pm$ 4 yr; height, 178.3 $\pm$ 6.1 cm; body mass, 83.7 $\pm$ 13.7 kg; peak oxygen uptake ($\dot{V}O_{2\text{peak}}$), 45.7 $\pm$ 9 ml·kg⁻¹·min⁻¹; unilateral 1-repetition maximum (1RM) leg press, 173.6 $\pm$ 37.1 kg (left leg), and 176.4 $\pm$ 33.7 kg (right leg)] completed this study. Participants were fully informed of all proce-

dures and potential risks involved before providing written consent. All procedures were approved by the Victoria University Human Research Ethics Committee.

Experimental Design

This study followed a within-subject, repeated-measures design. After familiarization and preliminary testing, participants completed three experimental trials in a randomized order. Each trial was separated by ~1 wk and participants were asked to maintain habitual diet and exercise habits during this period. Experimental trials were 1) RE performed alone (RE trial); 2) HIT cycling followed by RE (HIT + RE trial); or 3) work-matched MICT cycling followed by RE (MICT + RE trial).

Preliminary Testing

Preliminary testing was undertaken at least 1 wk before the first experimental trial to determine the lactate threshold (LT), $\dot{V}O_{2\text{peak}}$, and 1RM strength. Preliminary testing data was used to standardize relative exercise intensities during experimental trials with respect to the LT and 1RM. Endurance exercise intensities were standardized relative to the LT rather than $\dot{V}O_{2\text{peak}}$, as physiological responses are more consistent, independent of training status, when exercise is undertaken at an intensity relative to LT, compared with $\dot{V}O_{2\text{peak}}$ (6).

Graded exercise test. The participants' LT and peak power output (W_{peak}) were determined during a graded exercise test (GXT) to volitional exhaustion on an electromagnetically braked cycle ergometer (Lode, Groningen, The Netherlands). Before the GXT, a venous catheter was inserted into an antecubital forearm vein for subsequent blood sampling. The GXT comprised 4-min work stages interspersed with 30 s of passive recovery. Participants were required to maintain a pedaling cadence of 70 rpm during each work stage. The initial workload was set at 60, 90, or 120 W (to limit the number of stages to a maximum of 10, as determined during familiarization) and increased by 30 W for each subsequent stage until volitional exhaustion, defined as an inability to maintain a cadence >60 rpm. Venous blood samples (~1 ml) were obtained at rest and immediately following completion of each work stage. Whole blood samples were immediately analyzed in duplicate for lactate concentration using an automated analyzer (YSI 2300 STAT, Yellow Springs, OH). The LT was defined as the first workload that elicited a >1 mM increase in venous blood lactate concentration (23) and was calculated using Lactate-E version 2.0 software (66). The W_{peak} was determined as previously described (46).

Peak oxygen uptake ($\dot{V}O_{2\text{peak}}$) test. Immediately following the GXT, a 5-min active recovery was initiated at 20 W, after which participants cycled to volitional exhaustion at a workload corresponding to 105% of the W_{peak} achieved during the GXT. Participants were instructed to accelerate to a cadence of 90–100 rpm upon a 5-s countdown, and the test was terminated when a cadence >60 rpm was no longer possible. Expired gases were sampled every 15 s during this test component using automated gas analyzers (Moxus Modular $\dot{V}O_2$ System; AEI Technologies, Pittsburgh, PA). A similar protocol has previously been reported to elicit $\dot{V}O_{2\text{peak}}$ values no different to that determined during a ramp incremental test performed 5 min previously (71). The gas analyzers and pneumotach were calibrated before each test using known gas concentrations (21.0% O_2 and 0.04% CO_2 , 16.0% O_2 and 4.0% CO_2) and a 3-l calibration syringe, respectively. The individual $\dot{V}O_{2\text{peak}}$ was defined as the highest two consecutive 15-s values achieved during the test.

Maximal strength (1RM) testing. Maximal strength was determined during a series of 1RM unilateral leg press attempts using a plate-loaded 45° incline leg press (Hammer Strength Linear, Schiller Park, IL). After a standardized warm-up (5 and 3 repetitions for each leg at 50 and 70% estimated 1RM, respectively), single repetitions of increasing load were attempted for each leg until the maximal load possible for one repetition was determined. The test commenced with

the dominant leg, and a 1-min recovery was allowed before the other leg performed the equivalent 1RM attempt. Three minutes of recovery was allowed between 1RM attempts for the same leg. Participants completed each repetition with the heel placed at the bottom edge of the foot plate, and with a range of motion of 90° knee flexion/extension.

Exercise and Dietary Control

For 24 h before each experimental trial, participants were asked to refrain from strenuous and/or structured physical activity and were provided with a standardized diet. The daily energy intake (~149 kJ/kg; ~36 kcal/kg) comprised ~5.1 g/kg carbohydrate, ~1.3 g/kg protein, and ~1.2 g/kg fat. The percentage contribution of each macronutrient to the daily energy intake was 62% carbohydrate, 12% protein, and 26% fat. On the morning of each experimental trial, a standardized breakfast (~13 kJ/kg; ~3.1 kcal/kg) providing ~0.7 g/kg carbohydrate, ~0.1 g/kg protein, and <0.01 g/kg fat was ingested 90 min before the initial muscle biopsy. The percentage contribution of each macronutrient to the breakfast energy intake was 85% carbohydrate, 13% protein, and 2% fat.

Experimental Trial Overview

On the morning of an experimental trial (Fig. 1), participants reported to the laboratory at ~7:00 AM after a 10- to 12-h fast. A venous catheter was then inserted into an antecubital forearm vein for subsequent blood sampling, and a resting blood sample was obtained. Participants then ingested the standardized breakfast (described previously). For the RE trial, participants then rested quietly in the laboratory for 90 min before a resting biopsy was obtained and the RE protocol commenced (described subsequently). For the HIT + RE and MICT + RE trials, participants performed identical procedures as the RE trial; however, either the HIT or work-matched MICT protocol (both described subsequently) was completed 15 min before RE. After completion of RE, participants rested quietly in the laboratory for 3 h, and additional vastus lateralis biopsies were obtained 1 h and 3 h after completion of RE.

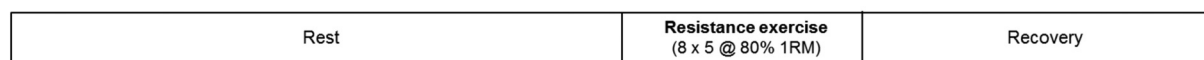
Cycling exercise. After a standardized warm-up (5 min at 75 W), participants commenced a bout of either HIT (10 × 2 min at 120% LT interspersed with 1 min of passive recovery) or work- and duration-matched MICT (30 min at 80% LT) (34) (see Fig. 1) on an electromagnetically braked cycle ergometer (Velotron; Racer-Mate, Seattle, WA). Heart rate (Polar FT1, Kempele, Finland) and rating of perceived exertion (RPE) (14) were obtained at regular intervals throughout exercise. Cycling was performed in the seated position and at a fixed cadence of 80 and 100 rpm for MICT and HIT, respectively. Following completion of endurance exercise, participants rested passively for 15 min before undertaking RE.

Resistance exercise. After a standardized warm-up (5 and 3 repetitions at 50 and 60% 1RM, respectively), participants completed 8 × 5 unilateral leg press repetitions on each leg at 80% of the 1RM. The biopsied leg first completed each set, and 1-min recovery was allowed before the opposing leg completed the equivalent set. Three minutes of recovery was allowed between sets for the same leg. Participants completed each repetition with the heel placed at the bottom edge of the foot plate, and with a range of motion of 90° knee flexion/extension. All participants successfully completed the identical resistance exercise protocol for each experimental trial, which was not influenced by prior HIT or MICT.

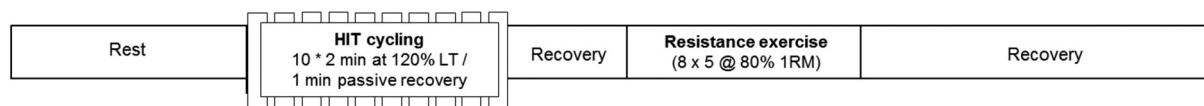
Blood sampling. Venous blood samples (~1 ml) were obtained at rest, immediately precycling, and after 10, 16, 22, 28, and 34 min of cycling. Additional blood samples were drawn after 2, 5, 10, and 15 min recovery from cycling, immediately following completion of RE, and after 2, 5, 10, 30, 60, 120, and 180 min of recovery from RE. Blood samples were aliquoted into a microtube (~1 ml) and immediately analyzed for glucose and lactate concentrations using an automated analyzer (YSI 2300, Yellow Springs, OH).

Muscle biopsies. Muscle biopsies were obtained from the middle portion of the vastus lateralis using the Bergström technique (11) modified with suction (36). After administration of local anesthesia (1% Xylocaine), three small incisions (~6 mm, ~1 cm apart) were made to the anesthetized skin in preparation for a series of muscle biopsies. Biopsies were obtained immediately before RE (REST),

A RE TRIAL



B HIT+RE TRIAL



C MICT+RE TRIAL

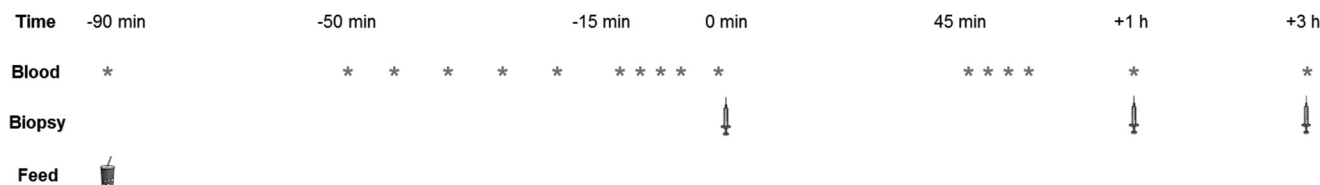
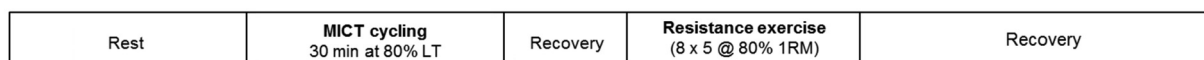


Fig. 1. Timelines for experimental trials. For the resistance exercise (RE) trial (A), participants completed the RE protocol alone. For the high-intensity interval training (HIT) + RE (B) and moderate-intensity continuous training (MICT) + RE (C) trials, RE was performed after a 15-min recovery from either the HIT or work-matched MICT cycling. Muscle biopsies were obtained from the vastus lateralis immediately before RE and either without (REST) or following (POST) prior endurance exercise and 1 h (RE + 1 h) and 3 h (RE + 3 h) after completion of RE. LT, lactate threshold; 1RM, one-repetition maximum.

after a preceding bout of HIT or work-matched MICT (POST), and 1 h (RE + 1 h) and 3 h (RE + 3 h) after RE. To minimize the number of biopsy samples taken from participants, the initial resting biopsy obtained from the RE trial was used as a basal reference for changes in signaling responses and gene expression for each trial (29). This allowed insight into the influence of prior HIT and MICT on responses within skeletal muscle before subsequent RE. All within-trial biopsies were obtained from the same leg, which was alternated for the subsequent trials. Muscle samples were blotted on filter paper to remove excess blood, immediately frozen in liquid nitrogen, and stored at -80°C until subsequent analysis.

Western Blotting

Approximately 15 mg of muscle tissue were homogenized in ice-cold lysis buffer (137 mM NaCl, 2.7 mM KCl, 1 mM MgCl_2 , 20 mM Tris, 1 mM EDTA, 1% Triton X-100, 10% glycerol, 5 mM Na pyrophosphate, 1 mM NaF, 1 $\mu\text{g}/\text{ml}$ leupeptin, 1 mM PMSF, 1 $\mu\text{g}/\text{ml}$ aprotinin, 1 mM Na_4VO_3 , 1 mM DTT, and 1 mM benzamide). Samples were rotated end-over-end for 1 h at 4°C and centrifuged at 15,000 g for 10 min at 4°C and the supernatant was collected. After determination of protein concentration (Bradford assay), the lysate was resuspended in 2 $\times$ Laemmli buffer (1.5 M Tris-HCl, 4% SDS, 20% glycerol, and 1.5% bromophenol blue) and stored at -80°C until subsequent analysis. Depending on the protein target, either 15 or 20 μg of protein were separated by SDS-PAGE using either 6, 8, or 12% acrylamide gels in a 1 $\times$ running buffer (25 mM Tris, 192 mM glycine, and 0.1% SDS) and then transferred to polyvinylidene fluoride (PVDF) membranes (Bio-Rad Laboratories, Hercules, CA) using a wet transfer for 90 min in 1 $\times$ transfer buffer (25 mM Tris, 192 mM glycine, 0.1% SDS, and 20% methanol). After transfer, membranes were blocked for 1 h at room temperature with 7.5% skim milk in 1 $\times$ TBST (200 mM Tris, 1.5 M NaCl, and 0.05% Tween 20), washed with 1 $\times$ TBST (5 $\times$ 5 min), and incubated with primary antibody solution (5% BSA, 0.05% Na Azide in 1 $\times$ TBST) overnight at 4°C . Primary antibodies for monoclonal p-mTOR^{Ser2448} (1:1,000; no. 5536), p-p70S6K^{Thr389} (1:1,000; no. 9234), p-4E-BP1^{Thr37/46} (1:1,000; no. 2855), p-AMPK^{Thr172} (1:1,000; no. 2535), p-rps6^{Ser235/236} (1:750; no. 4856), p-GSK-3 α/β ^{Ser21/9} (1:1,000; no. 9331), and polyclonal p-eEF2^{Thr56} (1:1,000; no. 2331), and p-ACC^{Ser79} (1:1,000, no. 3661) were from Cell Signaling Technology (Danvers, MA). The following morning, membranes were washed again with 1 $\times$ TBST and incubated with a secondary antibody (no. NEF812001EA; Perkin Elmer, Waltham, MA; 1:5,000 or 1:10,000 in 5% skim milk and 1 $\times$ TBST) for 1 h at room temperature. After being washed again with 1 $\times$ TBST, proteins were detected with chemiluminescence (Clarity Western ECL Substrate; Bio-Rad Laboratories) using a VersaDoc 4000 MP imaging system (Bio-Rad Laboratories) and quantified via densitometry (Image Lab 5.0; Bio-Rad Laboratories). All sample time points for each participant were run on the same gel and normalized to the total protein content of each lane using a modified Coomassie staining protocol (82). Briefly, after imaging, membranes were washed in 1 $\times$ TBST, stained with 0.1% Brilliant Blue R-350 (Sigma-Aldrich, St. Louis, MO) in 1:1 methanol/water solution for 2 min, de-stained in 1:5:4 ethanol/acetic acid/water solution for 1 min, rinsed briefly with water, and then air dried for ~ 1 h before imaging (see Fig. 4 for representative Western blot images of all proteins analyzed).

Real-Time Quantitative PCR

RNA extraction total. RNA was extracted from ~ 20 mg of muscle tissue using TRIzol reagent (Ambion, Austin, TX) according to the manufacturer's protocol. Muscle samples were homogenized in 500 μl of TRIzol reagent using a MagNA Lyser (Roche Diagnostics, Indianapolis, IN) for 20 s at 5,500 rpm. After resting for 5 min on ice, 50 μl of 1-bromo-3-chloropropane were added and the tube inverted for 30 s, before resting for 10 min at room temperature. The homog-

enate was centrifuged for 15 min at 13,000 rpm and the upper clear phase was collected. Then, 300 μl of isopropanol were added to the tube and inverted briefly to mix, before being stored overnight at -20°C to precipitate the RNA. After overnight incubation, the solution was centrifuged for 60 min at 13,000 rpm to pellet the RNA. The RNA pellet was washed twice by centrifuging in 75% ethanol for 15 min at 13,000 rpm, allowed to air dry, and then dissolved in 15 μl of nuclease-free water (NFW; Ambion). The quantity and quality of RNA were subsequently determined using a spectrophotometer (NanoDrop ND-1000; Thermo Scientific, Wilmington, DE). The purity of RNA was assessed using the ratio between the absorbance at 260 nm and absorbance at 280 nm (mean $\pm$ SD; 2.15 ± 0.37) and the ratio between the absorbance at 260 nm and absorbance at 230 nm (1.32 ± 0.51).

Reverse transcription. For mRNA analysis, first-strand cDNA was generated from 1 μg RNA in 20 μl reaction buffer using the high-capacity cDNA reverse transcriptase (RT) kit (Applied Biosystems) according to manufacturer's protocol, with each reaction comprising 2 μl of 10 $\times$ RT buffer, 0.8 μl of dNTP mix (100 mM), 2 μl of 10 $\times$ random primers, 1 μl of MultiScribe Reverse Transcriptase, 1 μl of RNase inhibitor, 3.2 μl of NFW, and 10 μl of RNA sample (diluted to 100 ng/ μl). Reverse transcription was then performed with the following conditions: 10 min at 25°C to anneal primers, 120 min at 37°C for the extension phase, and 5 min at 85°C . Following reverse transcription, samples were DNase-treated (Life Technologies, Carlsbad, CA) and cDNA was stored at -20°C until further analysis.

For miRNA analysis, first-strand cDNA was generated from 50 ng RNA in 15 μl reaction buffer with the TaqMan miRNA RT kit (no. 4366596; Applied Biosystems) and TaqMan miRNA-specific stem-loop primers (500 ng; no. 4427975; Applied Biosystems) using a modified multiplex protocol (55). Briefly, miRNA-specific primers were pooled and diluted in NFW to obtain a final dilution of 0.05 $\times$ each. Then, 6 μl of pooled primer solution were added to the reaction mix containing 0.3 μl of 100 mM dNTP, 3 μl of enzyme (50 U/ μl), 1.5 μl of 10 $\times$ RT buffer, 0.19 μl of RNase inhibitor (20 U/ μl), and 50 ng of RNA sample. Reverse transcription was then performed with the following conditions: 30 min at 16°C to anneal primers, 30 min at 42°C for the extension phase, and 5 min at 85°C . The cDNA was then stored at -20°C until further analysis.

Real-time quantitative PCR. Real-time PCR was performed using a Realplex² Mastercycler PCR system (Eppendorf, Hamburg, Germany) to measure mRNA levels of MuRF-1, Atrogin-1, Fox-O1, myostatin, TSC2, Rheb, PGC-1 α , and the reference genes cyclophilin, TBP, GAPDH, and $\beta 2\text{M}$. Each PCR reaction was performed in duplicate using a robotic pipetting machine (Epmotion 2100; Eppendorf) in a final reaction volume of 10 μl containing 5 μl of 2 $\times$ SYBR green (Bio-Rad Laboratories), 0.6 μl of PCR primers (diluted to 15 μM ; Sigma-Aldrich), 0.4 μl of NFW, and 4 μl of cDNA sample (diluted to 5 ng/ μl). Conditions for the PCR reactions were as follows: 3 min at 95°C , 40 cycles of 15 s at 95°C /1 min at 60°C , one cycle of 15 s at 95°C /15 s at 60°C , and a ramp for 20 min to 95°C . Each plate was briefly centrifuged before loading into the PCR machine. To compensate for variations in input RNA amounts and efficiency of the reverse transcription, mRNA data were normalized to the geometric mean of the two most stable housekeeping genes analyzed (cyclophilin and TBP), determined as previously described (60). Details of PCR primers used for all mRNA targets are shown in Table 1. Standard and melting curves were performed for all primers to confirm both primer efficiency and single product amplification, respectively.

A Stratagene MX3000 PCR system (Agilent Technologies, Santa Clara, CA) was used to measure the relative expression levels of miR-1, miR-133a, miR-378, miR-473, and RNU48. Each PCR reaction was performed in triplicate in a final reaction volume of 20 μl containing 7.5 μl of 2 $\times$ TaqMan Universal PCR Master Mix, no UNG (no. 4440040; Applied Biosystems), 0.5 μl of 1 $\times$ TaqMan Small RNA Assay (20 $\times$; no. 4427975; Applied Biosystem), 0.4 μl of NFW, and 4 μl of cDNA sample (diluted to 5 ng/ μl). Conditions for

Table 1. Details of PCR primers used for mRNA analysis

Gene	Forward Sequence	Reverse Sequence	NCBI Reference Sequence
MuRF-1	5'-CCTGAGAGCCATTGACTTTGG-3'	5'-CTTCCCTTCTGTGGACTCTTCCT-3'	NM_032588.3
Atrogin-1	5'-GCAGCTGAACAACATTGACATGAC-3'	5'-CAGCCTTGCATGATGTTTCACT-3'	NM_058229.3
Myostatin	5'-CCAGGAGAAGATGGGCTGAA-3'	5'-CAAGACCAAAATCCCTTCTGGAT-3'	NM_005259
Fox-O1	5'-TTGTTACATAGTCAGCTTG-3'	5'-TCACTTTCCTGCCCCAACAG-3'	NM_002015.3
TSC2	5'-CCGCAGCATCAGTGTGTC-3'	5'-CACTGGTGAGGGACGCTG-3'	NM_000548
Rheb	5'-TTTTTGGAACTCTTCTGCTAAAGAAA-3'	5'-AAGACTTGCCCTGTGAAGCTG-3'	NM_005614
PGC-1 α	5'-GGCAGAAGGCAATTGAAGAG-3'	5'-TCAAAACGGTCCCTCAGTTC-3'	NM_013261.3
TBP	5'-CAGTGACCCAGCAGCATCACT-3'	5'-AGGCCAAGCCCTGAGCGTAA-3'	M55654.1
Cyclophilin	5'-GTCAACCCACCGTGTCTTCTC-3'	5'-TTTCTGCTGTCTTTGGGACCTTG-3'	XM_011508410.1
GAPDH	5'-AAAGCGTGGCGGTGACTAAC-3'	5'-CGCCCAATACGACCAATCAGA-3'	NM_001256799.2
β 2M	5'-TGCTGTCTCCATGTTTGTATGATCT-3'	5'-TCTCTGCTCCGCACCTCTAAGT-3'	NM_004048.2

MuRF-1, muscle RING-finger 1; Fox-O1, forkhead-box O1; TSC2, tuberous sclerosis complex 2; Rheb, ras homologue enriched in brain; PGC-1 α , peroxisome proliferator-activated receptor- γ coactivator 1 α ; TBP, TATA binding protein; GAPDH, glyceraldehyde 3-phosphate dehydrogenase; β 2M, beta-2 microglobulin.

the PCR reactions were 10 min at 95°C, followed by 40 cycles of 15 s at 95°C, and 1 min at 60°C. Each plate was briefly centrifuged before being loaded into the PCR machine. To compensate for variations in input RNA amounts and efficiency of the reverse transcription, miRNA data were normalized to RNU48 expression. Details of PCR primers used for all miRNA targets are shown in Table 2. Standard curves were performed for all miRNA targets to confirm primer efficiency.

Muscle Glycogen Content

Approximately 2–3 mg of freeze-dried muscle tissue were powdered and dissected free of all visible nonmuscle tissue. Powdered muscle tissue was then extracted with 250 μ l of 2 M HCl, incubated at 95°C for 2 h (agitated gently every 20 min), and then neutralized with 750 μ l of 0.66 M NaOH. Glycogen content was subsequently assayed in triplicate via enzymatic analysis with flourimetric detection (45). Muscle glycogen values are expressed as millimoles per kilograms dry weight.

Statistical Analysis

The effect of trial on outcomes was evaluated via a two-way (trial $\times$ time) ANOVA with repeated-measures (SPSS, Version 21; IBM, New York, NY). We also used the magnitude-based approach to inferences using the standardized difference [effect size (ES)], as previously described (50). The magnitude of effects was defined according to thresholds suggested by Hopkins (50), where <0.2 = trivial, 0.2 – 0.6 = small, 0.6 – 1.2 = moderate, 1.2 – 2.0 = large, 2.0 – 4.0 = very large, and >4.0 = extremely large effects. Lacking information on the smallest meaningful effect for changes in protein phosphorylation and gene expression, we defined the threshold for the smallest worthwhile effect as an ES of 0.4, rather than the conventional threshold of 0.2 (50). Magnitude-based inferences about effects were made by qualifying the effects with probabilities that reflected the uncertainty in the magnitude of the true effect (9); 25–75%, “possibly”; 75–95%, “likely”; 95–99.5%, “very likely”; and $>99.5\%$, “most likely.” Effects that were deemed at least “likely” to be substantial (according to the overlap between the uncertainty in the magnitude of the true effect and the smallest worthwhile effect; Ref. 9) were reported. All magnitude-based inference data are provided in

Supplemental Tables 1 and 2 (Supplemental Material for this article is available online at the Journal website). Exact P values were also determined for each comparison, derived from paired (for within-trial comparisons) or unpaired (for between-trial comparisons) t -tests, with a Bonferroni correction applied to correct for multiple comparisons (SPSS, Version 21; IBM). Physiological (blood lactate, blood glucose, and heart rate) and psychological (RPE) responses to exercise are reported as mean values $\pm$ SD, whereas molecular (protein signaling and mRNA and RNA expression) and muscle glycogen data were log-transformed before analysis to reduce nonuniformity of error (50) and reported as mean between-condition percentage differences $\pm 90\%$ confidence limits (CL).

RESULTS

Venous Blood Lactate and Glucose Responses

Venous blood lactate levels were higher for HIT compared with MICT at all time points during cycling and recovery from cycling ($P < 0.05$; Table 3). Blood lactate was also higher for HIT + RE at 10, 30, and 60 min post-RE compared with RE (see Table 4; $P < 0.05$). Venous blood glucose was higher for HIT compared with MICT only after 34 min of cycling, and at 2, 5, and 10 min of recovery from cycling ($P < 0.05$; Table 3).

Heart Rate and RPE Responses

Both heart rate (mean $\pm$ SD, 171 ± 9 and 135 ± 12 beats/min, respectively) and RPE during cycling (16 ± 2 and 12 ± 2 arbitrary units, respectively) were higher during HIT compared with MICT ($P < 0.05$; Table 3).

Muscle Glycogen Content

There was a main effect of time for changes in muscle glycogen content ($P < 0.001$). Muscle glycogen content was unchanged across time following RE (Fig. 2); however, compared with REST, it was reduced at POST for both HIT + RE (mean difference $\pm 90\%$ CL, $50 \pm 23\%$; ES $\pm 90\%$ CL, -2.91 ± 1.86 ; $P = 0.022$) and MICT + RE ($40 \pm 11\%$; ES, -2.10 ± 0.76 ; $P = 0.004$). Compared with RE, muscle glycogen content was also lower for both HIT + RE and MICT + RE at RE + 1 h ($62 \pm 15\%$; ES, -4.01 ± 1.56 ; $P = 0.002$ and $43 \pm 19\%$; ES, -2.36 ± 1.38 ; $P = 0.014$, respectively) and RE + 3 h ($45 \pm 11\%$; ES, -2.46 ± 0.80 ; $P = 0.001$ and $31 \pm 18\%$; ES, -1.55 ± 1.07 ; $P = 0.033$, respectively). Muscle glycogen content was also lower for HIT + RE compared with MICT + RE at RE + 1 h ($33 \pm 22\%$; ES, -1.65 ± 1.32 ; $P = 0.034$).

Table 2. Details of primers used for microRNA analysis

Target	Assay Name	Product ID
miR-1	hsa-miR-1	002222
miR-133a	hsa-miR-133a-3p	002246
miR-378	hsa-miR-378	002243
miR-486	hsa-miR-486-5p	001278
RNU48	RNU48	001006

Table 3. Physiological and psychological (RPE) responses to HIT and work-matched MICT protocols

	Time, min										
	Rest	Pre	10	16	22	28	34	+2	+5	+10	+15
Lactate, mmol/l											
HIT	0.6 ± 0.2	0.8 ± 0.2	2.8 ± 1.0†	4.3 ± 1.5*†	5.3 ± 1.7*†	5.7 ± 1.9*†	6.1 ± 2.0*†	5.8 ± 1.8*†	5.6 ± 2.2†	4.9 ± 2.2†	4.1 ± 2.0†
MICT	0.6 ± 0.2	0.9 ± 0.2	1.6 ± 0.4†	2.1 ± 0.6	2.3 ± 0.7	2.4 ± 0.6	2.4 ± 0.7	2.2 ± 0.6	2.0 ± 0.6	1.6 ± 0.4	1.4 ± 0.2
Glucose, mmol/l											
HIT	4.4 ± 0.7	4.3 ± 1.9	3.4 ± 1.2	3.7 ± 0.7	4.1 ± 0.1	4.3 ± 0.5	4.6 ± 0.5†	5.0 ± 0.8†	4.8 ± 0.5†	4.6 ± 0.6†	4.3 ± 0.5
MICT	4.4 ± 0.3	4.6 ± 1.0	3.6 ± 0.6	3.4 ± 0.3	3.5 ± 0.4	3.7 ± 0.4	3.8 ± 0.6	4.0 ± 0.4	4.0 ± 0.5	4.0 ± 0.5	4.0 ± 0.5
Heart rate, beats/min											
HIT	—	63 ± 12	160 ± 9*†	168 ± 7*†	173 ± 9*†	176 ± 9*†	178 ± 8*†	—	—	—	—
MICT	—	62 ± 10	123 ± 9*	133 ± 12*	139 ± 13*	141 ± 12*	141 ± 12*	—	—	—	—
RPE, AU											
HIT	—	6 ± 0	14 ± 2*†	15 ± 2*†	16 ± 2*†	17 ± 2*†	18 ± 1*†	—	—	—	—
MICT	—	6 ± 0	10 ± 1*	11 ± 2*	12 ± 2*	12 ± 2*	13 ± 2*	—	—	—	—

Values are means ± SD. HIT, high-intensity interval training cycling; MICT, continuous cycling; RPE, rating of perceived exertion; AU, arbitrary units. * $P < 0.05$ vs. rest; † $P < 0.05$ vs. MICT at same time point.

Signaling Responses

p-AMPK^{Thr172}. Following RE, the phosphorylation of AMPK^{Thr172} was increased compared with REST at RE + 1 h ($137 \pm 162\%$; ES, 1.12 ± 0.83 ; $P = 0.033$) and RE + 3 h ($201 \pm 143\%$; ES, 1.43 ± 0.60 ; $P = 0.002$; Fig. 3A). Compared with REST, there were moderate effects for elevated AMPK phosphorylation at POST for HIT + RE ($107 \pm 163\%$; ES, 0.94 ± 0.94 ; $P = 0.144$) and MICT + RE ($118 \pm 166\%$; ES, 1.01 ± 0.88 ; $P = 0.074$). A moderate effect was also seen for higher AMPK phosphorylation at RE + 1 h for MICT + RE compared with RE ($59 \pm 61\%$; ES, 0.61 ± 0.48 ; $P = 0.085$). There were no differences in AMPK phosphorylation between HIT + RE and MICT + RE at any time point.

p-ACC^{Ser79}. There were main effects of time ($P < 0.001$), trial ($P = 0.002$), and an interaction effect ($P < 0.001$), for ACC^{Ser79} phosphorylation. Compared with REST, the phosphorylation of ACC^{Ser79} was lower for RE at RE + 1 h ($60 \pm 24\%$; ES, -1.15 ± 0.70 ; $P = 0.017$; Fig. 3B) and RE + 3 h ($82 \pm 28\%$; ES, -2.15 ± 1.53 ; $P = 0.040$). ACC phosphorylation was higher at POST compared with REST for both HIT + RE ($530 \pm 145\%$; ES, 2.29 ± 0.28 ; $P < 0.001$) and MICT + RE ($451 \pm 274\%$; ES, 2.13 ± 0.60 ; $P = 0.002$). ACC phosphorylation was higher for HIT + RE compared with RE both at RE + 1 h ($133 \pm 96\%$; ES, 1.06 ± 0.50 ; $P = 0.038$) and + 3 h ($458 \pm 215\%$; ES, 2.14 ± 0.47 ; $P = 0.043$). Moreover, ACC phosphorylation was higher for HIT + RE compared with MICT + RE at RE + 1 h ($161 \pm$

218% ; ES, 1.19 ± 0.95 ; $P = 0.046$) and RE + 3 h ($161 \pm 121\%$; ES, 1.2 ± 0.56 ; $P = 0.005$).

p-mTOR^{Ser2448}. There was a main effect of trial for mTOR^{Ser2448} phosphorylation ($P = 0.035$). The phosphorylation of mTOR^{Ser2448} was unchanged over time by RE; however, it was higher at POST for HIT + RE compared with both REST ($105 \pm 34\%$; ES, 0.84 ± 0.19 ; $P = 0.011$; Fig. 3C) and MICT + RE ($44 \pm 21\%$; ES, 0.67 ± 0.44 ; $P = 0.014$). At RE + 1 h, mTOR phosphorylation was higher for HIT + RE compared with MICT + RE ($128 \pm 67\%$; ES, 0.83 ± 0.29 ; $P = 0.003$). mTOR phosphorylation was also moderately higher at RE + 3 h for HIT + RE compared with both RE ($91 \pm 84\%$; ES, 0.75 ± 0.49 ; $P = 0.148$) and MICT + RE ($138 \pm 285\%$; ES, 1.01 ± 1.18 ; $P = 0.150$).

p-p70S6K^{Thr389}. There was a main effect of time for p70S6K^{Thr389} phosphorylation ($P = 0.001$). RE increased p70S6K^{Thr389} phosphorylation above REST values at RE + 1 h ($171 \pm 95\%$; ES, 1.39 ± 0.48 ; $P = 0.002$; Fig. 3D) and RE + 3 h ($65 \pm 64\%$; ES, 0.70 ± 0.47 ; $P = 0.025$). At POST, small-to-moderate elevations in p70S6K phosphorylation were noted for HIT + RE ($77 \pm 55\%$; ES, 0.80 ± 0.43 ; $P = 0.154$) and MICT + RE ($60 \pm 67\%$; ES, 0.66 ± 0.57 ; $P = 0.178$) compared with REST. There were no differences in p70S6K phosphorylation between HIT + RE and MICT + RE at any time point.

p-rps6^{Ser235/236}. There was a main effect of time for changes in rps6^{Ser235/236} phosphorylation ($P = 0.030$). RE increased

Table 4. Venous blood lactate and glucose responses following resistance exercise performed alone (RE) and when performed after either HIT (HIT + RE) or work-matched MICT (MICT + RE)

	Time, min							
	End	+2	+5	+10	+30	+60	+120	+180
Lactate, mmol/l								
RE	1.7 ± 0.7	1.7 ± 0.8	1.5 ± 0.6	1.2 ± 0.4	0.8 ± 0.2	0.6 ± 0.2	0.6 ± 0.2	0.5 ± 0.2
HIT + RE	2.2 ± 0.9	2.3 ± 0.9	2.1 ± 0.8	1.7 ± 0.7†	1.1 ± 0.4*	0.9 ± 0.4*	0.7 ± 0.4	0.6 ± 0.1
MICT + RE	1.7 ± 1.7	1.7 ± 0.7	1.5 ± 0.3	1.4 ± 0.4	0.9 ± 0.2	0.8 ± 0.2	0.7 ± 0.1	0.6 ± 0.1
Glucose, mmol/l								
RE	4.4 ± 0.3	4.1 ± 0.6	4.4 ± 0.2	4.3 ± 0.3	4.3 ± 0.3	4.4 ± 0.6	4.4 ± 0.3	4.5 ± 0.2
HIT + RE	4.1 ± 0.4	4.2 ± 0.3	4.2 ± 0.2	4.1 ± 0.4	4.2 ± 0.3	4.2 ± 0.3	4.3 ± 0.3	4.3 ± 0.3
MICT + RE	4.3 ± 0.3	4.3 ± 0.3	4.2 ± 0.4	4.2 ± 0.4	4.4 ± 0.2	4.3 ± 0.2	4.3 ± 0.3	4.4 ± 0.2

Values are means ± SD. RE, resistance exercise; HIT + RE, high-intensity interval training cycling and resistance exercise; MICT + RE, continuous cycling and resistance exercise. * $P < 0.05$ vs. RE at same time point.

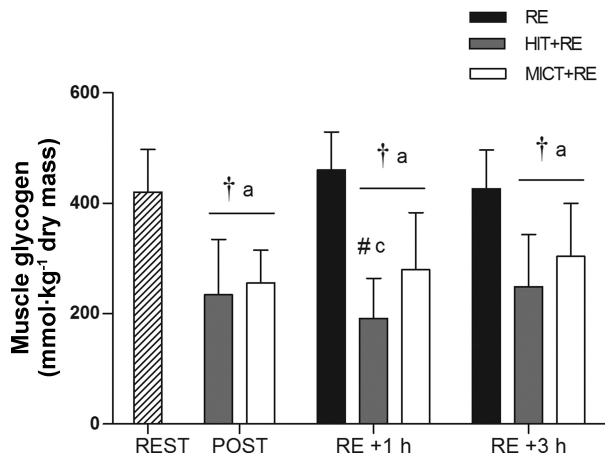


Fig. 2. Muscle glycogen content of the vastus lateralis at REST, after a single bout of endurance exercise (POST), and 1 h (RE + 1 h) and 3 h (RE + 3 h) after a bout of resistance exercise (RE) performed alone or 15 min after a bout of either high-intensity interval training (HIT) + RE or moderate-intensity continuous training (MICT) + RE. Data presented are means $\pm$ SD. $\dagger P < 0.05$ vs. RE at same time point; $\# P < 0.05$ vs. MICT + RE at same time point. Substantially different at same time point vs. *RE and *MICT + RE.

rps6^{Ser235/236} phosphorylation above REST values at RE + 1 h ($215 \pm 155\%$; ES, 1.21 ± 0.50 ; $P = 0.029$; Fig. 3E) and + 3 h ($203 \pm 226\%$; ES, 1.17 ± 0.73 ; $P = 0.018$). At POST, rps6 phosphorylation was higher for HIT + RE compared with REST ($153 \pm 116\%$; ES, 0.98 ± 0.47 ; $P = 0.023$). There were no differences in rps6 phosphorylation between HIT + RE and MICT + RE at any time point.

p-eEF2^{Thr56}. The phosphorylation status of eEF2^{Thr56} was unchanged over time by RE (Fig. 3F). However, there were large effect for reduced eEF2 phosphorylation at POST for both MICT + RE ($37 \pm 23\%$; ES, -1.87 ± 1.47 ; $P = 0.035$) and HIT + RE ($34 \pm 33\%$; ES, -1.72 ± 1.98 ; $P = 0.144$) compared with REST. Moreover, compared with RE, there was a moderate effect for lower eEF2 phosphorylation at RE + 1 h for both HIT + RE ($16 \pm 10\%$; ES, -0.71 ± 0.49 ; $P = 0.050$) and MICT + RE ($21 \pm 24\%$; ES, -0.99 ± 1.22 ; $P = 0.179$). There were no differences in eEF2 phosphorylation between HIT + RE and MICT + RE at any time point.

p-GSK-3 β ^{Ser9}. There was a main effect of time for GSK-3 β ^{Ser9} phosphorylation ($P = 0.021$). RE increased GSK-3 β ^{Ser9} phosphorylation above REST values at + 1 h ($130 \pm 87\%$; ES, 0.70 ± 0.31 ; $P = 0.002$; Fig. 3G), returning to baseline at RE + 3 h. There were no between-trial differences in GSK-3 β phosphorylation at any time point.

p-4E-BP1^{Thr36/47}. The phosphorylation status of 4E-BP1^{Thr36/47} was unchanged across time by RE and was not different between trials at any time point (Figs. 3H and 4).

mRNA Responses

MuRF-1 mRNA. There was a main effect of time for MuRF-1 mRNA expression ($P = 0.048$). MuRF-1 expression was not altered by RE at RE + 3 h compared with REST (Fig. 5A); however, there were similar small increases in MuRF-1 expression at RE + 3 h for both MICT + RE ($535 \pm 464\%$; ES, 0.33 ± 0.20 ; $P = 0.016$) and HIT + RE ($585 \pm 684\%$; ES, 0.52 ± 0.64 ; $P = 0.170$) compared with RE. There were no differences in MuRF-1 expression between HIT + RE and MICT + RE at any time point.

Atrogin-1 (MaFbx) mRNA. Atrogin-1 expression was not altered by RE at RE + 3 h compared with REST (Fig. 5B), and was not different between trials at any timepoint.

Fox-O1 mRNA. Fox-O1 expression was unaltered by RE at RE + 3 h compared with REST (Fig. 5C). There was a small increase in Fox-O1 mRNA expression at RE + 3 h for HIT + RE compared with MICT + RE ($62 \pm 58\%$; ES, 0.34 ± 0.25 ; $P = 0.035$).

Myostatin mRNA. Myostatin expression was unaltered by RE at RE + 3 h compared with REST (Fig. 5D). At POST, myostatin expression was greater for HIT + RE compared with MICT + RE ($53 \pm 18\%$; ES, 0.60 ± 0.30 ; $P = 0.044$).

PGC-1 α mRNA. There was a main effect of time for PGC-1 α mRNA expression ($P = 0.016$). PGC-1 α expression was unchanged after RE (Fig. 5E); however, there were comparable small increases in PGC-1 α expression at RE + 3 h for both MICT + RE ($604 \pm 403\%$; ES, 0.59 ± 0.42 ; $P = 0.033$) and HIT + RE ($788 \pm 878\%$; ES, 0.54 ± 0.54 ; $P = 0.098$) compared with RE. There were no differences in PGC-1 α expression between HIT + RE and MICT + RE at any time point.

Rheb/TSC2 mRNA. mRNA expression of Rheb or TSC2 was not altered by RE, and was not different between trials at any time point (Fig. 5, F and G).

miRNA Responses

miR-1. There was a main effect of time for miR-1 expression ($P = 0.043$). However, miR-1 expression was not altered by RE, and was not substantially different between trials at any time point (Fig. 6A).

miR-133a. There was a main effect of time for miR-133a expression ($P = 0.035$). RE decreased miR-133a expression relative to REST at RE + 3 h ($27 \pm 14\%$; ES, -0.38 ± 0.23 ; $P = 0.026$; Fig. 6B). For the HIT + RE trial, miR-133a expression was lower at POST compared with REST ($40 \pm 13\%$; ES, -0.63 ± 0.27 ; $P = 0.014$), and there were small-to-moderate reductions in miR-133a expression at RE + 1 h compared with both RE ($35 \pm 28\%$; ES, -0.53 ± 0.50 ; $P = 0.068$) and MICT + RE ($43 \pm 30\%$; ES, -0.68 ± 0.61 ; $P = 0.053$). There were no between-condition differences in miR-133a expression at RE + 3 h.

miR-378. RE did not alter miR-378 expression relative to REST (Fig. 6C). However, there were moderate reductions in miR-378 expression at RE + 1 h for HIT + RE compared with both RE ($37 \pm 26\%$; ES, -0.79 ± 0.70 ; $P = 0.052$) and MICT + RE ($42 \pm 28\%$; ES, -0.94 ± 0.81 ; $P = 0.046$). There were no between-condition differences in miR-378 expression at RE + 3 h.

miR-486. There was a main effect of trial for miR-486 expression ($P = 0.048$). RE did not alter miR-486 expression relative to REST (Fig. 6D). However, miR-486 expression was reduced at RE + 1 h for HIT + RE compared with both RE ($36 \pm 17\%$; ES, -1.02 ± 0.59 ; $P = 0.040$) and MICT + RE ($38 \pm 15\%$; ES, -1.10 ± 0.56 ; $P = 0.015$). There were no between-condition differences in miR-486 expression at RE + 3 h.

DISCUSSION

We aimed to compare the effects of concurrent exercise incorporating either HIT or MICT on anabolic and catabolic molecular responses in skeletal muscle, compared with RE performed alone. The major findings were that 1) prior HIT and MICT induced similar muscle glycogen depletion and

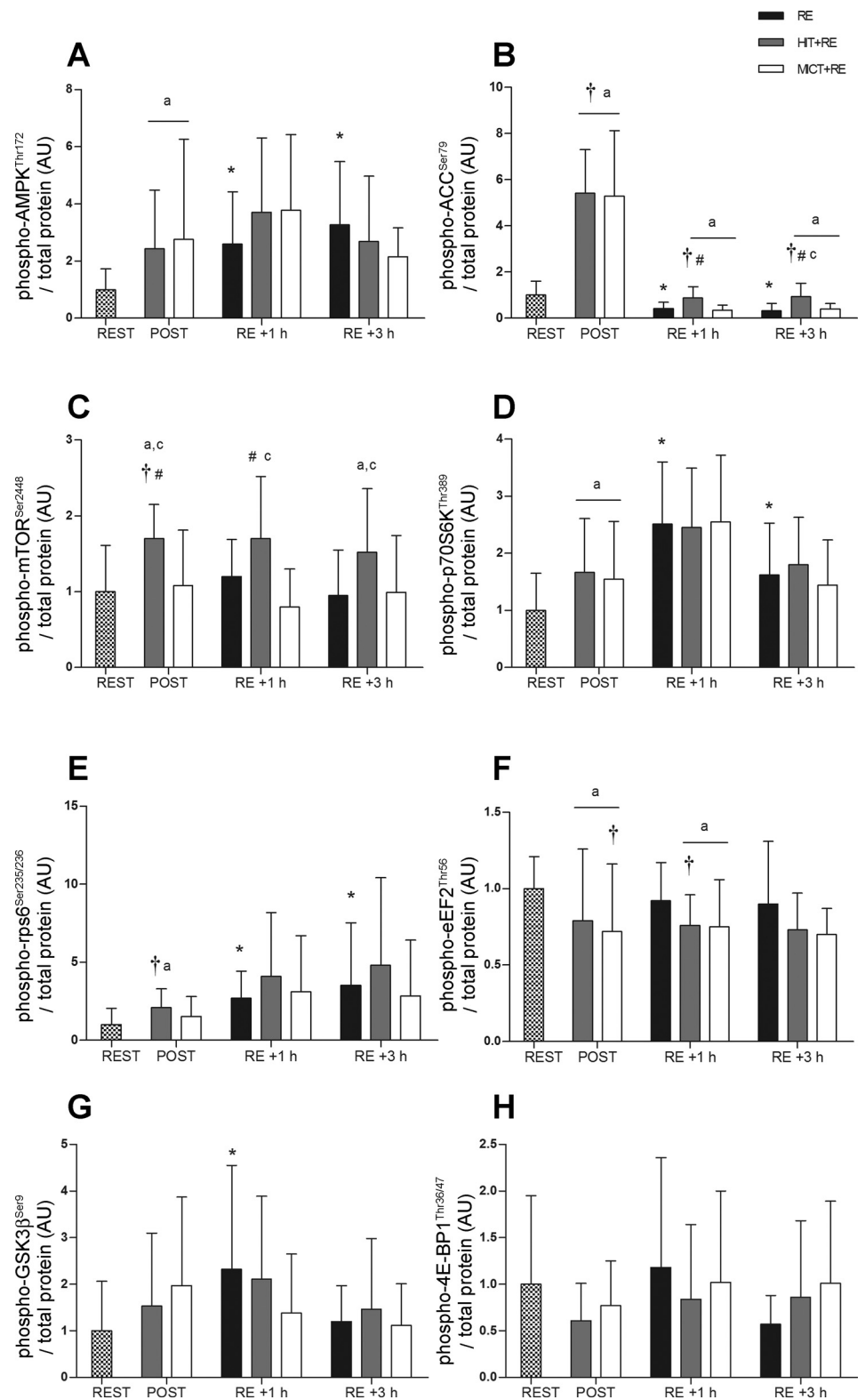


Fig. 3. Phosphorylation of AMPK^{Thr172} (A), ACC^{Ser79} (B), mTOR^{Ser2448} (C), p70S6K^{Thr389} (D), rps6^{Ser235/236} (E), eEF2^{Thr56} (F), GSK-3β^{Ser9} (G), and 4E-BP1^{Thr36/47} (H) at REST, after a single bout of endurance exercise (POST), and 1 h (RE + 1 h) and 3 h (RE + 3 h) after a bout of resistance exercise (RE) either performed alone 15 min after a bout of either high-intensity interval training (HIT; HIT + RE) or moderate-intensity continuous training (MICT; MICT + RE). Data presented are means $\pm$ SD and expressed relative to the REST value for the RE trial. AU, arbitrary units. * P < 0.05 vs. REST; † P < 0.05 vs. RE at same time point; # P < 0.05 vs. MICT + RE at same time point. Substantially different at same time point vs. *RE and †MICT + RE.

increased ACC phosphorylation before commencing a subsequent RE bout; 2) despite these residual effects, mTORC1 signaling was generally not compromised during early recovery from concurrent exercise compared with RE alone; 3) concurrent exercise incorporating HIT and MICT similarly increased MuRF-1 mRNA expression compared with RE

alone; and 4) prior HIT, but not MICT, reduced the expression of miRNAs implicated in exercise-induced skeletal muscle adaptations after subsequent RE, compared with RE performed alone.

Interference to mTORC1 signaling has been investigated as a mechanism contributing to attenuated RE adaptations during

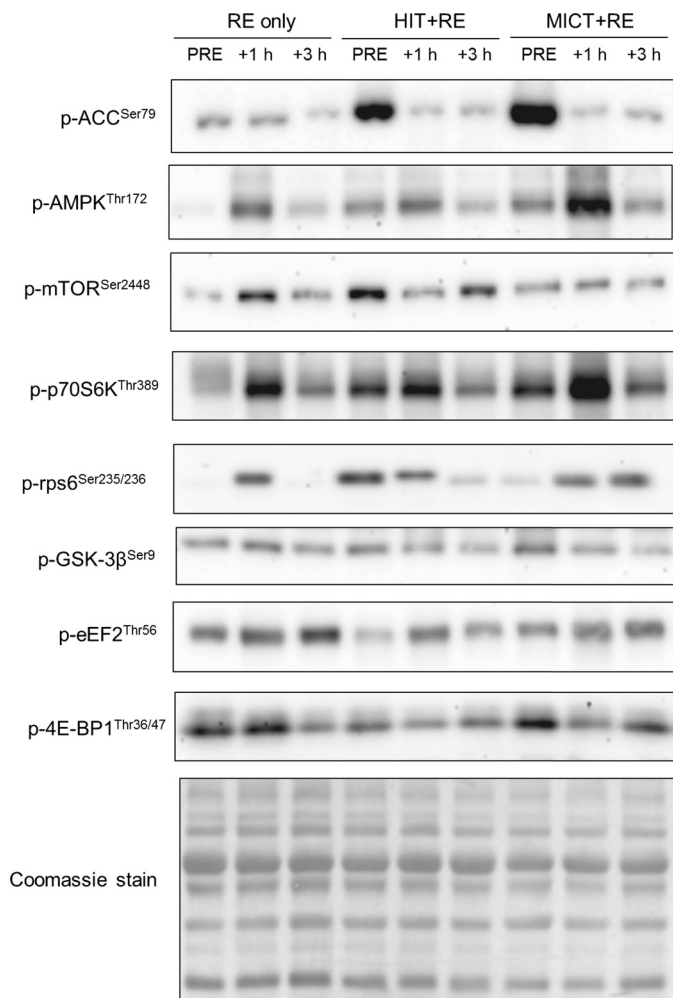


Fig. 4. Representative Western blots for the phosphorylation (p-) of signaling proteins at REST, after a single bout of endurance exercise (POST), and 1 h (RE + 1 h) and 3 h (RE + 3 h) after a bout of resistance exercise (RE) performed alone or 15 min after a bout of either high-intensity interval training (HIT; HIT + RE) or moderate-intensity continuous training (MICT; MICT + RE).

concurrent training in humans (1, 29, 57). However, whether higher endurance exercise intensities, previously shown to further increase AMPK activity in skeletal muscle compared with lower intensity endurance exercise (70), are associated with greater interference to mTORC1 signaling following concurrent exercise has not been examined. In the present study, the RE-induced phosphorylation responses of several mTORC1 pathway intermediates were generally not compromised when RE was performed after either HIT or MICT. For example, the phosphorylation of p70S6K, which reportedly correlates with load-induced hypertrophy in both rodent (4) and human (78) models, was similar between conditions at 1 h post-RE. We observed few differences in postexercise signaling responses between concurrent exercise and single-mode RE, although eEF2 phosphorylation was lower at RE + 1 h for HIT + RE compared with RE alone, and mTOR phosphorylation was lower at RE + 1 h for MICT + RE compared with HIT + RE and RE alone. These observations align with others showing that mTORC1 signaling and rates of MPS following a single bout of concurrent exercise are not attenuated com-

pared with RE performed alone (1, 29, 57). For example, performing 30 min of cycling at 70% $\dot{V}O_{2\max}$ 15 min after RE does not compromise the postexercise phosphorylation of mTOR, p70S6K, and eEF2 compared with RE performed alone (1). Moreover, rates of MPS following RE are not compromised by the addition of either 20 min of MICT cycling at 55% peak aerobic power (29) or 90 min at 60% peak aerobic power (17). These data add support to the notion that concurrent exercise appears not to interfere with mTORC1 signaling compared with single-mode RE, at least within the early postexercise recovery period (<3 h).

Rather than inhibit anabolic responses, the endurance exercise protocols in the present study were themselves clearly sufficient to promote mTORC1 signaling. The HIT protocol was particularly effective for increasing mTOR and rps6 phosphorylation compared with MICT, whereas both protocols similarly increased p70S6K phosphorylation at POST. These data align with evidence that endurance exercise is sufficient to induce anabolic responses in skeletal muscle, including increased mTORC1 signaling and rates of MPS, at least in relatively training-unaccustomed individuals (28, 61). For example, mTOR/p70S6K phosphorylation and rates of mixed MPS are increased following 1 h of cycling at 65–70% $\dot{V}O_{2\max}$ (61), and high (60% $W_{\max}$ for 30 min)- but not low (30% $W_{\max}$ for 60 min)-intensity continuous cycling induces sustained (24–28 h postexercise) increases in myofibrillar MPS (28). The importance of training status in dictating the specificity of early postexercise responses must also be considered when interpreting these data. For example, mTOR phosphorylation is preferentially increased following RE, but not endurance exercise, in training-acclimated individuals (80). Similarly, both myofibrillar and mitochondrial MPS are similarly increased by resistance and endurance exercise in a training-unaccustomed state, whereas RE preferentially stimulates increased myofibrillar MPS after training (83). It therefore appears likely that concurrent exercise modulates early postexercise molecular responses over time, potentially progressing towards an attenuation of anabolic responses with concurrent compared with single-mode RE. Indeed, the greater increase in p70S6K phosphorylation observed after a single bout of concurrent exercise compared with single-mode RE is abolished after 5 wk of training (37). The participants in the present study were not accustomed to the specific exercise protocols employed during the experimental trials, which may have influenced the ability of endurance exercise to modulate anabolic responses in skeletal muscle. Further work is required to determine the influence of training status and the potential progression of interference to anabolic responses and adaptations in human skeletal muscle with concurrent training.

Muscle glycogen depletion, a modulator of AMPK activity (89), has been suggested as a potential mechanism by which endurance exercise may attenuate anabolic responses in skeletal muscle following RE. While early work (24) suggested Akt phosphorylation was attenuated in human skeletal muscle when RE was performed in a glycogen-depleted state, more recent evidence suggests that muscle glycogen availability does not modulate mTORC1 signaling or rates of MPS following RE (16). The HIT and MICT protocols in the present study induced similar muscle glycogen depletion, so that glycogen content was comparably reduced at POST for the HIT + RE and MICT + RE trials compared with REST. Although we

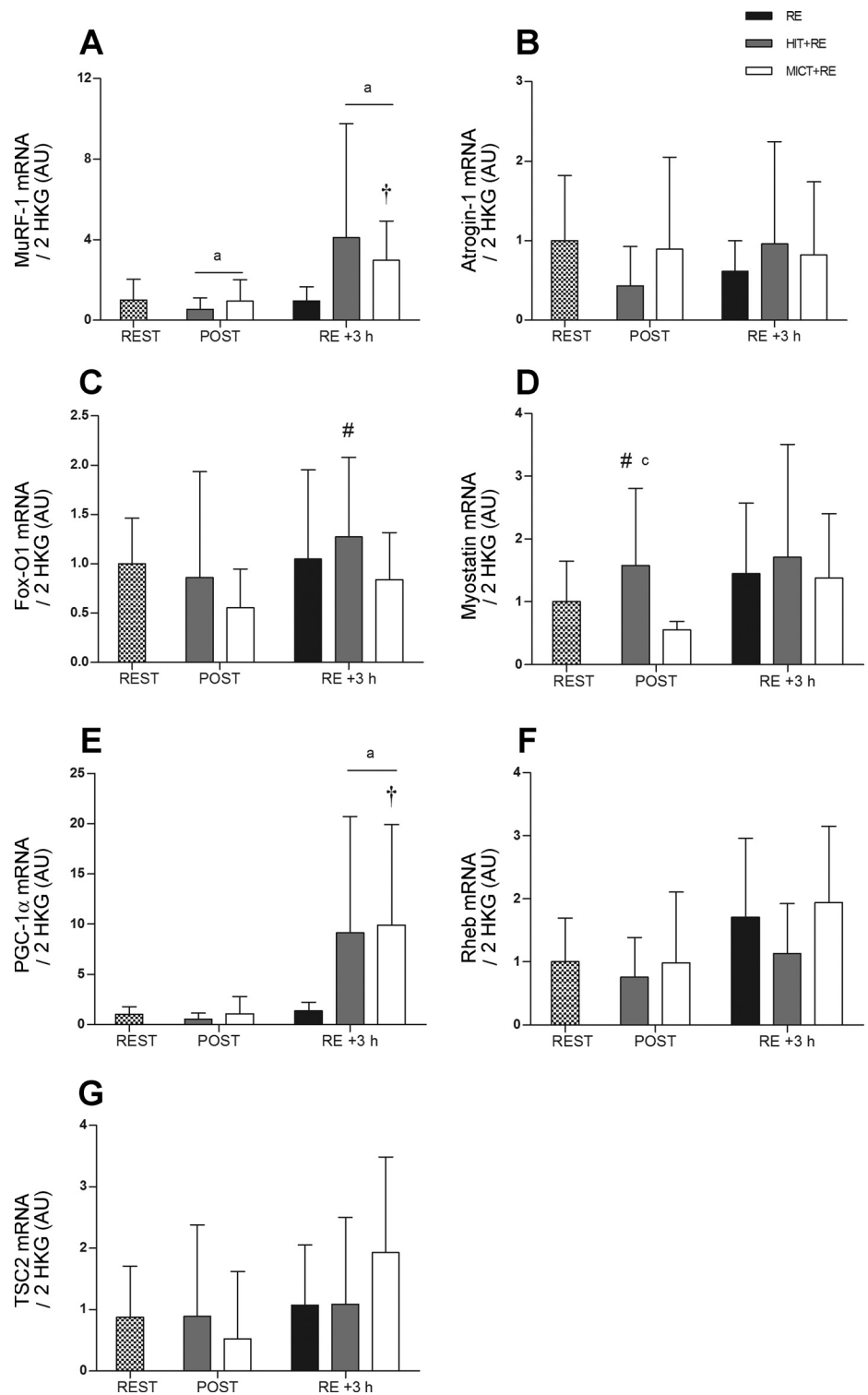


Fig. 5. mRNA expression of MuRF-1 (A), Atrogin-1 (B), Fox-O1 (C), myostatin (D), PGC-1 α (E), Rheb (F), and TSC2 (G) relative to the geometric mean of cyclophilin and TBP expression at REST, after a single bout of endurance exercise (POST) and 1 h (RE + 1 h) and 3 h (RE + 3 h) after a bout of resistance exercise (RE) performed alone or 15 min after a bout of either high-intensity interval training (HIT; HIT + RE) or moderate-intensity continuous training (MICT; MICT + RE). Data presented are means $\pm$ SD and expressed relative to the REST value for the RE trial. $\dagger P < 0.05$ vs. RE at same time point; $\# P < 0.05$ vs. MICT + RE at same time point. Substantially different at same time point vs. a RE and c MICT + RE.

initially hypothesized HIT would induce greater glycogen depletion, based on intensity-dependent glycogen depletion during continuous exercise (40), our findings align with previous work showing similar muscle glycogen depletion, at least at the whole muscle level, following constant or variable-intensity cycling (77) and running (8) matched for total work. Mirroring these responses were the comparable increases in ACC phosphorylation at POST with prior HIT or MICT, which

may suggest similar increases in AMPK activity (68). This is in agreement with Bartlett et al. (8), who observed similarly increased AMPK phosphorylation following work-matched HIT and MICT running. Although HIT and MICT induced similar glycogen depletion at POST, muscle glycogen content was $\sim 33\%$ lower at RE + 1 h for HIT + RE compared with MICT + RE. We noted a similar pattern for ACC phosphorylation, which was similarly elevated at RE + 1 h and RE +

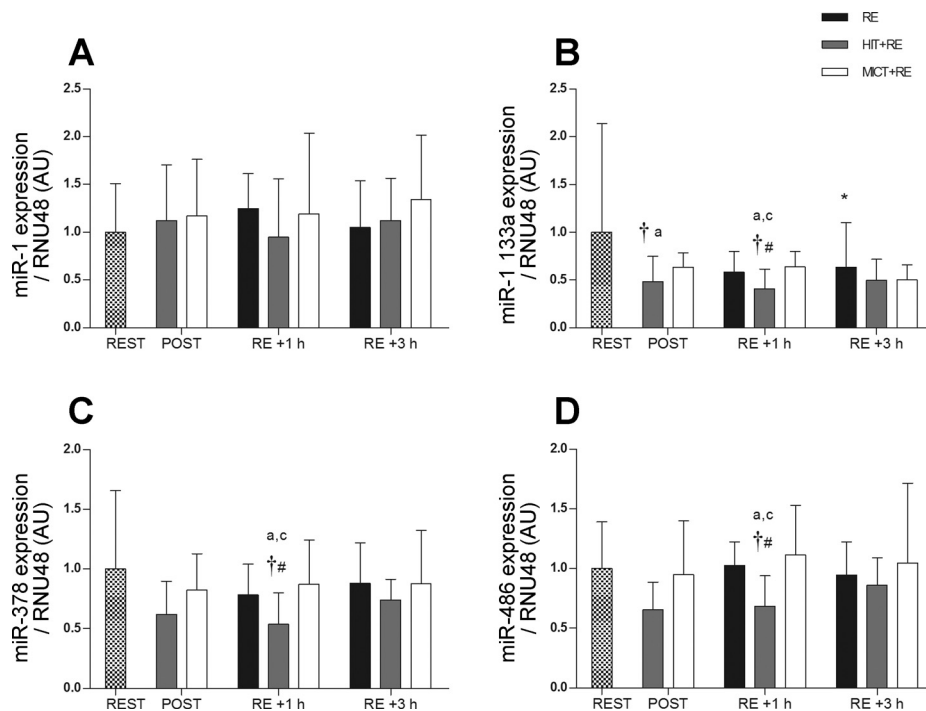


Fig. 6. Expression levels of miR-1 (A), miR-133a (B), miR-378 (C), and miR-486 (D) relative to RNU48 expression immediately at REST, after a single bout of endurance exercise (POST), and 1 h (RE + 1 h) and 3 h (RE + 3 h) after a bout of resistance exercise (RE) performed alone or 15 min after a bout of either high-intensity interval training (HIT; HIT + RE) or moderate-intensity continuous training (MICT; MICT + RE). Data presented are means $\pm$ SD and expressed relative to the REST value for the RE trial. * P < 0.05 vs. REST; † P < 0.05 vs. RE at same time point; # P < 0.05 vs. MICT + RE at same time point. Substantially different at same time point vs. ^aRE and ^cMICT + RE.

3 h (~161%) for HIT + RE compared with MICT + RE. Nevertheless, our data support the notion that muscle glycogen availability, at least in the ranges reported here, does not modulate exercise-induced mTORC1 signaling in human skeletal muscle and does not appear to mediate any interference effect within the time course examined.

In agreement with some (21, 30, 52), but not all (1, 15), studies we observed increased AMPK phosphorylation following RE. Potentially, the training status of our participants may have influenced these results, as AMPK phosphorylation has been shown to be increased by RE in participants unaccustomed to the specific exercise protocol (20, 30), yet AMPK activity is unchanged after RE in resistance-trained subjects (2). Despite the increased AMPK phosphorylation induced by RE, this was not associated with increases in ACC phosphorylation, a downstream target of AMPK and a marker of its activity (44, 68), which instead decreased over time for the RE trial. There appears, therefore, to be an apparent disconnect between the AMPK and ACC phosphorylation responses for the RE trial, for which we can offer no clear explanation. Dissociations between AMPK and ACC phosphorylation have been reported in human skeletal muscle with increases in exercise intensity (18) and during prolonged, submaximal exercise (86). Nevertheless, given that ACC phosphorylation may provide a better indication of AMPK activity compared with AMPK phosphorylation (68), we have primarily used the ACC phosphorylation responses when interpreting the effects of the exercise protocols on AMPK activity.

In addition to altering postexercise responses governing MPS, another mechanism by which concurrent exercise might mediate the interference effect is by exacerbating rates of protein degradation (2). MuRF-1 and Atrogin-1 are E3 ubiquitin ligases implicated in the tagging and subsequent degradation of contractile proteins via the ubiquitin proteasome system (12). In the present study, RE had little effect on either

MuRF-1 or Atrogin-1 gene expression relative to baseline. However, relative to RE alone, MuRF-1 mRNA abundance was similarly increased at RE + 3 h when RE was performed after HIT or MICT. While data regarding ubiquitin ligase expression after concurrent exercise incorporating divergent endurance training intensities are scarce, increased MuRF-1 mRNA expression has been shown in separate studies when RE is preceded by either HIT (2, 69) or MICT cycling (59). The present data lend support to the notion that concurrent exercise increases the expression of markers of protein degradation compared with single-mode RE and extend current knowledge by suggesting this response is unrelated to the endurance exercise intensity employed. However, given that increased postexercise ubiquitin ligase activity may be necessary to facilitate skeletal muscle remodeling by removing damaged proteins and/or providing amino acid substrates for incorporation into newly synthesized proteins (73), the significance of increased MuRF-1 expression following concurrent exercise is unclear.

We also investigated additional molecular mechanisms implicated in the regulation of exercise-induced skeletal muscle adaptations. miRNAs have been associated with the regulation of skeletal muscle mass (49), and their expression levels are altered following both single bouts of exercise (31, 72, 91) and short-term exercise training (26, 72). However, no studies have investigated whether miRNA expression is altered following concurrent compared with single-mode RE. The myomiRs miR-1 and miR-133a can target members of the IGF-1/Akt pathway *in vitro*, including IGF-1 (35), the IGF-1 receptor (35), as well as HSP70 (54). This suggests they may play a role in regulating muscle hypertrophy, although this requires experimental validation *in vivo*. Expression of both miR-1 and miR-133a is reduced during functional overload in mice (62), and miR-1 expression is reduced 1 h after a single bout of RE with amino acid ingestion (31), whereas a single bout of

endurance exercise increases miR-1 expression (72). Nonetheless, contrary to previous observations (31), we failed to observe a decrease in miR-1 expression after RE or any between-trial differences in this response following HIT + RE or MICT + RE. When performed alone, RE was sufficient to decrease miR-133a expression at RE + 3 h compared with baseline, while prior HIT further reduced miR-133a expression at REST and at RE + 1 h compared with RE. These observations suggest that concurrent HIT + RE may promote a more favorable postexercise anabolic response compared with RE alone by reducing miR-133a expression, thereby alleviating any potential miR-133a-mediated repression of members of the IGF-1/Akt signaling pathway.

Increased expression of miR-378 has been implicated in myogenesis in C2C12 cells by repressing MyoR, a negative upstream regulator of the transcription factor MyoD (39). Low responders to RE-induced lean mass gain after 12 wk of RE training show reduced basal miR-378 expression, whereas miR-378 expression is unchanged in those classified as high responders (26). The change in miR-378 expression has also been correlated ($r^2 = 0.52$) with lean mass gain after RE training (26), which suggests maintenance of miR-378 expression may be necessary for promoting RE-induced muscle growth. We observed that concurrent exercise incorporating HIT, but not MICT, reduced miR-378 expression at RE + 1 h relative to that induced by RE alone. However, whether this early postexercise reduction in miR-378 expression displays a similar relationship with lean mass gain following short-term training, as do changes in basal miR-378 expression (26), remains to be determined.

miR-486 has been linked to the regulation of skeletal muscle mass by targeting components of the Akt pathway, including phosphatase and tensin homologue (PTEN), an upstream inhibitor of Akt, and by negatively regulating the transcription factor Fox-O1 (88), a mediator of ubiquitin ligase expression and subsequently protein degradation (74). Increased postexercise miR-486 expression, when repeated over time, may therefore be favourable for promoting anabolism in skeletal muscle by alleviating PTEN-mediated repression of Akt and suppressing Fox-O1-mediated ubiquitin ligase expression. In the present study, miR-486 expression was decreased at RE + 1 h for the HIT + RE trial compared with RE alone, potentially suggesting reduced miR-486-mediated inhibition of PTEN and Fox-O1. Whether this early postexercise response translates into downstream effects on PTEN and Fox-O1 protein levels is, however, difficult to determine within the time course of the present study. We observed increased, rather than decreased, Fox-O1 mRNA expression for the HIT + RT trial compared with MICT + RT. This still does not discount the possibility that miR-486 may mediate its effects by either degrading Fox-O1 mRNA at a later timepoint or by repressing its translation. Further work using extended time courses is required to investigate whether these early postexercise changes in miRNA expression with concurrent exercise are associated with changes in predicted target mRNA and/or protein levels. Taken together, these data suggest concurrent exercise incorporating HIT, but not MICT, alters the postexercise expression of miRNAs purported to play a role in exercise-induced adaptations in skeletal muscle. However, further work is required to determine the functional roles of these miRNAs in human

skeletal muscle with exercise and their subsequent effects on skeletal muscle adaptations after prolonged training.

Limitations

As reported in similar studies investigating molecular interference with concurrent training (1, 21, 22, 37, 57, 59, 69), we used the phosphorylation of components of the mTORC1 pathway as a proxy for the activation of protein synthesis. However, a direct coupling between mTORC1 signaling and protein synthesis rates does not always exist (3), nor does greater phosphorylation of mTOR pathway intermediates always equal a greater protein synthesis response (25). Moreover, even functional measures, such as rates of MPS after a single bout of RE, do not correlate with muscle hypertrophy after 16 wk of RE (64). Further work is required to define the molecular events that mediate chronic phenotypic adaptations to training and potentially their progression during a training program. Such information is likely to also provide further mechanistic insight into the concurrent interference effect (38). We also conducted all muscle analyses using mixed whole muscle homogenate, which may have masked any fiber-type specific differences in measured postexercise responses, including glycogen content, signaling responses and gene expression (65). Indeed, while the HIT and MICT protocols induced similar whole muscle glycogen depletion and markers of AMPK activity, there might be fiber-type specific differences in these responses (77). Thus whether the prior endurance exercise protocols altered signaling responses and/or gene expression in a fibre-type specific manner following subsequent RE remains to be determined.

We employed a preexercise feeding protocol in an attempt to mimic typical exercise/nutritional scenarios, where exercise is not performed in a fasted state, an approach used in similar studies (37, 57, 59, 69). This feeding protocol was purposely low in protein content (~ 0.1 g/kg), thereby limiting the potential impact of amino acid provision on inducing anabolic signaling responses. While we believe our feeding approach better mimics typical training scenarios, this may limit our ability to compare our results to previous studies examining molecular responses in skeletal muscle to concurrent exercise performed in the fasted state (1, 2, 21, 22, 29).

Conclusions

To the best of our knowledge, this is the first investigation to directly compare the effects of HIT and work-matched MICT on mTORC1 signaling and miRNA expression in human skeletal muscle when performed concurrently with RE. We show that although HIT and MICT similarly reduce muscle glycogen content, and increased markers of AMPK activity before a subsequent RE bout, postexercise mTORC1 signaling is generally not compromised compared with RE performed alone. However, concurrent exercise incorporating HIT or MICT similarly increased MuRF-1 expression relative to single-mode RE, potentially indicating increased rates of protein degradation. Concurrent exercise incorporating HIT, but not MICT, also reduced the postexercise expression of miRNAs linked to the regulation of skeletal muscle hypertrophy. Taken together, our results suggest that HIT is a potent stimulus for inducing anabolic responses in skeletal muscle after concurrent exercise, including increased mTOR and rps6 phosphorylation, and for

altering the expression of miRNAs implicated in adaptations to RE, at least in relatively training-unaccustomed subjects.

Perspectives and Significance

This study directly compared the incorporation of either HIT or MICT into a concurrent training bout on postexercise molecular responses in skeletal muscle, compared with RE performed alone. Together, these data suggest the following:

- 1) Combining RE with HIT, rather than MICT, may provide a greater anabolic stimulus in skeletal muscle by promoting mTORC1 signaling and reducing expression levels of miRNAs purported to negatively regulate muscle hypertrophy signaling.
- 2) The incorporation of HIT into concurrent training programs may therefore be more favourable, rather than inhibitory, for the accretion of muscle mass and strength.
- 3) Further work is required to link the modulation of these early postexercise anabolic responses to phenotypic training adaptations, as well as the efficacy of incorporating either HIT or MICT into long-term concurrent training programs.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

J.J.F., D.J.B., and N.K.S. conception and design of research; J.J.F., D.J.B., and N.K.S. performed experiments; J.J.F., E.Z., and A.P.R. analyzed data; J.J.F., D.J.B., E.Z., A.P.R., and N.K.S. interpreted results of experiments; J.J.F. prepared figures; J.J.F. drafted manuscript; J.J.F., D.J.B., E.Z., A.P.R., and N.K.S. edited and revised manuscript; J.J.F., D.J.B., E.Z., A.P.R., and N.K.S. approved final version of manuscript.

REFERENCES

1. Apro W, Wang L, Ponten M, Blomstrand E, Sahlin K. Resistance exercise induced mTORC1 signaling is not impaired by subsequent endurance exercise in human skeletal muscle. *Am J Physiol Endocrinol Metab* 305: E22–E32, 2013.
2. Apro W, Moberg M, Hamilton DL, Ekblom B, van Hall G, Holmberg HC, Blomstrand E. Resistance exercise-induced S6K1 kinase activity is not inhibited in human skeletal muscle despite prior activation of AMPK by high-intensity interval cycling. *Am J Physiol Endocrinol Metab* 308: E470–E481, 2015.
3. Atherton PJ, Etheridge T, Watt PW, Wilkinson D, Selby A, Rankin D, Smith K, Rennie MJ. Muscle full effect after oral protein: time-dependent concordance and discordance between human muscle protein synthesis and mTORC1 signaling. *Am J Clin Nutr* 92: 1080–1088, 2010.
4. Baar K, Esser K. Phosphorylation of p70(S6k) correlates with increased skeletal muscle mass following resistance exercise. *Am J Physiol Cell Physiol* 276: C120–C127, 1999.
5. Balabinis CP, Psarakis CH, Moukas M, Vassiliou MP, Behrakis PK. Early phase changes by concurrent endurance and strength training. *J Strength Cond Res* 17: 393–401, 2003.
6. Baldwin J, Snow RJ, Febbraio MA. Effect of training status and relative exercise intensity on physiological responses in men. *Med Sci Sports Exerc* 32: 1648–1654, 2000.
7. Bartlett JD, Close GL, MacLaren DP, Gregson W, Drust B, Morton JP. High-intensity interval running is perceived to be more enjoyable than moderate-intensity continuous exercise: implications for exercise adherence. *J Sports Sci* 29: 547–553, 2011.
8. Bartlett JD, Hwa Joo C, Jeong TS, Louhelainen J, Cochran AJ, Gibala MJ, Gregson W, Close GL, Drust B, Morton JP. Matched work high-intensity interval and continuous running induce similar increases in PGC-1 α mRNA, AMPK, p38, and p53 phosphorylation in human skeletal muscle. *J Appl Physiol* 112: 1135–1143, 2012.
9. Batterham AM, Hopkins WG. Making meaningful inferences about magnitudes (Online). *Sportscience*. <http://sportsci.org/jour/05/ambwgh.htm> [14 Aug 2012].
10. Bell GJ, Syrotuik D, Martin TP, Burnham R, Quinney HA. Effect of concurrent strength and endurance training on skeletal muscle properties and hormone concentrations in humans. *Eur J Appl Physiol* 81: 418–427, 2000.
11. Bergstrom J. Muscle electrolytes in man. *Scand J Clin Lab Invest* 68: 1–110, 1962.
12. Bodine SC, Latres E, Baumhueter S, Lai VK, Nunez L, Clarke BA, Poueymirou WT, Panaro FJ, Na E, Dharmarajan K, Pan ZQ, Valenzuela DM, DeChiara TM, Stitt TN, Yancopoulos GD, Glass DJ. Identification of ubiquitin ligases required for skeletal muscle atrophy. *Science* 294: 1704–1708, 2001.
13. Bodine SC, Stitt TN, Gonzalez M, Kline WO, Stover GL, Bauerlein R, Zlotchenko E, Scrimgeour A, Lawrence JC, Glass DJ, Yancopoulos GD. Akt/mTOR pathway is a crucial regulator of skeletal muscle hypertrophy and can prevent muscle atrophy in vivo. *Nat Cell Biol* 3: 1014–1019, 2001.
14. Borg G. Physical training. 3. Perceived exertion in physical work. *Lakartidningen* 67: 4548–4557, 1970.
15. Camera DM, Edge J, Short MJ, Hawley JA, Coffey VG. Early time course of Akt phosphorylation after endurance and resistance exercise. *Med Sci Sports Exerc* 42: 1843–1852, 2010.
16. Camera DM, West DW, Burd NA, Phillips SM, Garnham AP, Hawley JA, Coffey VG. Low muscle glycogen concentration does not suppress the anabolic response to resistance exercise. *J Appl Physiol* 113: 206–214, 2012.
17. Carrithers JA, Carroll CC, Coker RH, Sullivan DH, Trappe TA. Concurrent exercise and muscle protein synthesis: implications for exercise countermeasures in space. *Aviat Space Environ Med* 78: 457–462, 2007.
18. Chen ZP, Stephens TJ, Murthy S, Canny BJ, Hargreaves M, Witters LA, Kemp BE, McConell GK. Effect of exercise intensity on skeletal muscle AMPK signaling in humans. *Diabetes* 52: 2205–2212, 2003.
19. Chtara M, Chaouachi A, Levin GT, Chaouachi M, Chamari K, Amri M, Laursen PB. Effect of concurrent endurance and circuit resistance training sequence on muscular strength and power development. *J Strength Cond Res* 22: 1037–1045, 2008.
20. Coffey VG, Zhong Z, Shield A, Canny BJ, Chibalin AV, Zierath JR, Hawley JA. Early signaling responses to divergent exercise stimuli in skeletal muscle from well-trained humans. *FASEB J* 20: 190–192, 2006.
21. Coffey VG, Jemiolo B, Edge J, Garnham AP, Trappe SW, Hawley JA. Effect of consecutive repeated sprint and resistance exercise bouts on acute adaptive responses in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 297: R1441–R1451, 2009.
22. Coffey VG, Pilegaard H, Garnham AP, O'Brien BJ, Hawley JA. Consecutive bouts of diverse contractile activity alter acute responses in human skeletal muscle. *J Appl Physiol* 106: 1187–1197, 2009.
23. Coyle EF, Coggan AR, Hemmert MK, Walters TJ. Glycogen usage and performance relative to lactate threshold. *Med Sci Sports Exerc* 16: 120–121, 1984.
24. Creer A, Gallagher P, Slivka D, Jemiolo B, Fink W, Trappe S. Influence of muscle glycogen availability on ERK1/2 and Akt signaling after resistance exercise in human skeletal muscle. *J Appl Physiol* 99: 950–956, 2005.
25. Crozier SJ, Kimball SR, Emmert SW, Anthony JC, Jefferson LS. Oral leucine administration stimulates protein synthesis in rat skeletal muscle. *J Nutr* 135: 376–382, 2005.
26. Daviden PK, Gallagher IJ, Hartman JW, Tarnopolsky MA, Dela F, Helge JW, Timmons JA, Phillips SM. High responders to resistance exercise training demonstrate differential regulation of skeletal muscle microRNA expression. *J Appl Physiol* (1985) 110: 309–417, 2011.
27. Derave W, Hansen BF, Lund S, Kristiansen S, Richter EA. Muscle glycogen content affects insulin-stimulated glucose transport and protein kinase B activity. *Am J Physiol Endocrinol Metab* 279: E947–E955, 2000.
28. Di Donato DM, West DW, Churchward-Venne TA, Breen L, Baker SK, Phillips SM. Influence of aerobic exercise intensity on myofibrillar and mitochondrial protein synthesis in young men during early and late

- postexercise recovery. *Am J Physiol Endocrinol Metab* 306: E1025–E1032, 2014.
29. Donges CE, Burd NA, Duffield R, Smith GC, West DW, Short MJ, Mackenzie R, Plank LD, Shepherd PR, Phillips SM, Edge JA. Concurrent resistance and aerobic exercise stimulates both myofibrillar and mitochondrial protein synthesis in sedentary middle-aged men. *J Appl Physiol* 112: 1992–2001, 2012.
 30. Dreyer HC, Fujita S, Cadenas JG, Chinkes DL, Volpi E, Rasmussen BB. Resistance exercise increases AMPK activity and reduces 4E-BP1 phosphorylation and protein synthesis in human skeletal muscle. *J Physiol* 576: 613–624, 2006.
 31. Drummond MJ, McCarthy JJ, Fry CS, Esser KA, Rasmussen BB. Aging differentially affects human skeletal muscle microRNA expression at rest and after an anabolic stimulus of resistance exercise and essential amino acids. *Am J Physiol Endocrinol Metab* 295: E1333–E1340, 2008.
 32. Drummond MJ, Fry CS, Glynn EL, Dreyer HC, Dhanani S, Timmerman KL, Volpi E, Rasmussen BB. Rapamycin administration in humans blocks the contraction-induced increase in skeletal muscle protein synthesis. *J Physiol* 587: 1535–1546, 2009.
 33. Dudley GA, Djamil R. Incompatibility of endurance- and strength-training modes of exercise. *J Appl Physiol* (1985) 59: 1446–1451, 1985.
 34. Edge J, Bishop D, Goodman C. The effects of training intensity on muscle buffer capacity in females. *Eur J Appl Physiol* 96: 97–105, 2006.
 35. Elia L, Contu R, Quintavalle M, Varrone F, Chimentì C, Russo MA, Cimino V, De Marinis L, Frustaci A, Catalucci D, Condorelli G. Reciprocal regulation of microRNA-1 and insulin-like growth factor-1 signal transduction cascade in cardiac and skeletal muscle in physiological and pathological conditions. *Circulation* 120: 2377–2385, 2009.
 36. Evans WJ, Phinney SD, Young VR. Suction applied to a muscle biopsy maximizes sample size. *Med Sci Sports Exerc* 14: 101–102, 1982.
 37. Fernandez-Gonzalo R, Lundberg TR, Tesch PA. Acute molecular responses in untrained and trained muscle subjected to aerobic and resistance exercise training versus resistance training alone. *Acta Physiol (Oxf)* 209: 283–294, 2013.
 38. Fyfe JJ, Bishop DJ, Stepto NK. Interference between concurrent resistance and endurance exercise: molecular bases and the role of individual training variables. *Sports Med* 44: 743–762, 2014.
 39. Gagan J, Dey BK, Laver R, Yan Z, Dutta A. MicroRNA-378 targets the myogenic repressor MyoR during myoblast differentiation. *J Biol Chem* 286: 19431–19438, 2011.
 40. Gollnick PD, Piehl K, Saltin B. Selective glycogen depletion pattern in human muscle fibres after exercise of varying intensity and at varying pedalling rates. *J Physiol* 241: 45–57, 1974.
 41. Gormley SE, Swain DP, High R, Spina RJ, Dowling EA, Kotipalli US, Gandrakota R. Effect of intensity of aerobic training on VO₂max. *Med Sci Sports Exerc* 40: 1336–1343, 2008.
 42. Grieco CR, Swain DP, Colberg SR, Dowling EA, Baskette K, Zarrabi L, Gandrakota R, Kotipalli U, Sechrist SR, Somma CT. Effect of intensity of aerobic training on insulin sensitivity/resistance in recreationally active adults. *J Strength Cond Res* 27: 2270–2276, 2013.
 43. Guo H, Ingolia NT, Weissman JS, Bartel DP. Mammalian microRNAs predominantly act to decrease target mRNA levels. *Nature* 466: 835–840, 2010.
 44. Ha J, Daniel S, Broyles SS, Kim KH. Critical phosphorylation sites for acetyl-CoA carboxylase activity. *J Biol Chem* 269: 22162–22168, 1994.
 45. Harris RC, Hultman E, Nordesjo LO. Glycogen, glycolytic intermediates and high-energy phosphates determined in biopsy samples of musculus quadriceps femoris of man at rest. Methods and variance of values. *Scand J Clin Lab Invest* 33: 109–120, 1974.
 46. Hawley JA, Noakes TD. Peak power output predicts maximal oxygen uptake and performance time in trained cyclists. *Eur J Appl Physiol Occup Physiol* 65: 79–83, 1992.
 47. Helgerud J, Rodas G, Kemi OJ, Hoff J. Strength and endurance in elite football players. *Int J Sports Med* 32: 677–682, 2011.
 48. Hickson RC. Interference of strength development by simultaneously training for strength and endurance. *Eur J Appl Physiol Occup Physiol* 45: 255–263, 1980.
 49. Hitachi K, Tsuchida K. Role of microRNAs in skeletal muscle hypertrophy. *Front Physiol* 4: 408, 2013.
 50. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc* 41: 3–13, 2009.
 51. Kimball SR. Interaction between the AMP-activated protein kinase and mTOR signaling pathways. *Med Sci Sports Exerc* 38: 1958–1964, 2006.
 52. Koopman R, Zorenc AH, Gransier RJ, Cameron-Smith D, van Loon LJ. Increase in S6K1 phosphorylation in human skeletal muscle following resistance exercise occurs mainly in type II muscle fibers. *Am J Physiol Endocrinol Metab* 290: E1245–E1252, 2006.
 53. Kraemer WJ, Patton JF, Gordon SE, Harman EA, Deschenes MR, Reynolds K, Newton RU, Triplett NT, Dziados JE. Compatibility of high-intensity strength and endurance training on hormonal and skeletal muscle adaptations. *J Appl Physiol* 78: 976–989, 1995.
 54. Kukreti H, Amuthavalli K, Harikumar A, Sathiyamoorthy S, Feng PZ, Anantharaj R, Tan SL, Lokireddy S, Bonala S, Sriram S, McFarlane C, Kambadur R, Sharma M. Muscle-specific microRNA1 (miR1) targets heat shock protein 70 (HSP70) during dexamethasone-mediated atrophy. *J Biol Chem* 288: 6663–6678, 2013.
 55. Le Carre J, Lamon S, Leger B. Validation of a multiplex reverse transcription and preamplification method using TaqMan(R) MicroRNA assays. *Front Genet* 5: 413, 2014.
 56. Leveritt M, Abernethy PJ, Barry BK, Logan PA. Concurrent strength and endurance training. A review. *Sports Med* 28: 413–427, 1999.
 57. Lundberg TR, Fernandez-Gonzalo R, Gustafsson T, Tesch PA. Aerobic exercise alters skeletal muscle molecular responses to resistance exercise. *Med Sci Sports Exerc* 44: 1680–1688, 2012.
 58. Lundberg TR, Fernandez-Gonzalo R, Gustafsson T, Tesch PA. Aerobic exercise does not compromise muscle hypertrophy response to short-term resistance training. *J Appl Physiol* 114: 81–89, 2013.
 59. Lundberg TR, Fernandez-Gonzalo R, Tesch PA. Exercise-induced AMPK activation does not interfere with muscle hypertrophy in response to resistance training in men. *J Appl Physiol* (1985) 116: 611–620, 2014.
 60. Mane VP, Heuer MA, Hillyer P, Navarro MB, Rabin RL. Systematic method for determining an ideal housekeeping gene for real-time PCR analysis. *J Biomol Tech* 19: 342–347, 2008.
 61. Mascher H, Ekblom B, Rooyackers O, Blomstrand E. Enhanced rates of muscle protein synthesis and elevated mTOR signalling following endurance exercise in human subjects. *Acta Physiol (Oxf)* 202: 175–184, 2011.
 62. McCarthy JJ, Esser KA. MicroRNA-1 and microRNA-133a expression are decreased during skeletal muscle hypertrophy. *J Appl Physiol* (1985) 102: 306–313, 2007.
 63. McCarthy JP, Pozniak MA, Agre JC. Neuromuscular adaptations to concurrent strength and endurance training. *Med Sci Sports Exerc* 34: 511–519, 2002.
 64. Mitchell CJ, Churchward-Venne TA, Parise G, Bellamy L, Baker SK, Smith K, Atherton PJ, Phillips SM. Acute post-exercise myofibrillar protein synthesis is not correlated with resistance training-induced muscle hypertrophy in young men. *PLoS One* 9: e89431, 2014.
 65. Murphy RM, Lamb GD. Important considerations for protein analyses using antibody based techniques: down-sizing Western blotting up-sizes outcomes. *J Physiol* 591: 5823–5831, 2013.
 66. Newell J, Higgins D, Madden N, Cruickshank J, Einbeck J, McMillan K, McDonald R. Software for calculating blood lactate endurance markers. *J Sports Sci* 25: 1403–1409, 2007.
 67. Olesen J, Küllerich K, Pilegaard H. PGC-1 α -mediated adaptations in skeletal muscle. *Pflügers Arch* 460: 153–162, 2010.
 68. Park SH, Gammon SR, Knippers JD, Paulsen SR, Rubink DS, Winder WW. Phosphorylation-activity relationships of AMPK and acetyl-CoA carboxylase in muscle. *J Appl Physiol* (1985) 92: 2475–2482, 2002.
 69. Pugh JK, Faulkner SH, Jackson AP, King JA, Nimmo MA. Acute molecular responses to concurrent resistance and high-intensity interval exercise in untrained skeletal muscle. *Physiol Rep* 3: e12364, 2015.
 70. Rose AJ, Bisiani B, Vistisen B, Kiens B, Richter EA. Skeletal muscle eEF2 and 4EBP1 phosphorylation during endurance exercise is dependent on intensity and muscle fiber type. *Am J Physiol Regul Integr Comp Physiol* 296: R326–R333, 2009.
 71. Rossiter HB, Kowalchuk JM, Whipp BJ. A test to establish maximum O₂ uptake despite no plateau in the O₂ uptake response to ramp incremental exercise. *J Appl Physiol* 100: 764–770, 2006.
 72. Russell AP, Lamon S, Boon H, Wada S, Guller I, Brown EL, Chibalin AV, Zierath JR, Snow RJ, Stepto N, Wadley GD, Akimoto T. Regulation of miRNAs in human skeletal muscle following acute endurance exercise and short-term endurance training. *J Physiol* 591: 4637–4653, 2013.
 73. Sanchez AM, Candau RB, Bernardi H. FoxO transcription factors: their roles in the maintenance of skeletal muscle homeostasis. *Cell Mol Life Sci* 71: 1657–71, 2014.

74. Sandri M, Sandri C, Gilbert A, Skurk C, Calabria E, Picard A, Walsh K, Schiaffino S, Lecker SH, Goldberg AL. Foxo transcription factors induce the atrophy-related ubiquitin ligase atrogin-1 and cause skeletal muscle atrophy. *Cell* 117: 399–412, 2004.
75. Sillanpää E, Laaksonen DE, Hakkinen A, Karavirta L, Jensen B, Kraemer WJ, Nyman K, Häkkinen K. Body composition, fitness, and metabolic health during strength and endurance training and their combination in middle-aged and older women. *Eur J Appl Physiol* 106: 285–96, 2009.
76. Silva RF, Cadore EL, Kothe G, Guedes M, Alberton CL, Pinto SS, Trindade G, Kruel LF. Concurrent training with different aerobic exercises. *Int J Sports Med* 33: 627–634, 2012.
77. Suriano R, Edge J, Bishop D. Effects of cycle strategy and fibre composition on muscle glycogen depletion pattern and subsequent running economy. *Br J Sports Med* 44: 443–448, 2010.
78. Terzis G, Georgiadis G, Stratakis G, Vogiatzis I, Kavouras S, Manta P, Mascher H, Blomstrand E. Resistance exercise-induced increase in muscle mass correlates with p70S6 kinase phosphorylation in human subjects. *Eur J Appl Physiol* 102: 145–152, 2008.
79. Tjønnå AE, Lee SJ, Rognmo O, Stolen TO, Bye A, Haram PM, Loennechen JP, Al-Share QY, Skogvoll E, Slørdahl SA, Kemi OJ, Najjar SM, Wisløff U. Aerobic interval training versus continuous moderate exercise as a treatment for the metabolic syndrome: a pilot study. *Circulation* 118: 346–354, 2008.
80. Vissing K, McGee SL, Farup J, Kjølhed T, Vendelbo MH, Jessen N. Differentiated mTOR but not AMPK signaling after strength vs endurance exercise in training-acclimated individuals. *Scand J Med Sci Sports* 23: 355–366, 2011.
81. Wang L, Mascher H, Psilander N, Blomstrand E, Sahlin K. Resistance exercise enhances the molecular signaling of mitochondrial biogenesis induced by endurance exercise in human skeletal muscle. *J Appl Physiol* 111: 1335–1344, 2011.
82. Welinder C, Ekblad L. Coomassie staining as loading control in Western blot analysis. *J Proteome Res* 10: 1416–1419, 2011.
83. Wilkinson SB, Phillips SM, Atherton PJ, Patel R, Yarasheski KE, Tarnopolsky MA, Rennie MJ. Differential effects of resistance and endurance exercise in the fed state on signalling molecule phosphorylation and protein synthesis in human muscle. *J Physiol* 586: 3701–3717, 2008.
84. Wilson JM, Marin PJ, Rhea MR, Wilson SM, Loenneke JP, Anderson JC. Concurrent training: a meta-analysis examining interference of aerobic and resistance exercises. *J Strength Cond Res* 26: 2293–2307, 2012.
85. Wisløff U, Støylen A, Loennechen JP, Bruvold M, Rognmo O, Haram PM, Tjønnå AE, Helgerud J, Slørdahl SA, Lee SJ, Videm V, Bye A, Smith GL, Najjar SM, Ellingsen Ø, Skjaerpe T. Superior cardiovascular effect of aerobic interval training versus moderate continuous training in heart failure patients: a randomized study. *Circulation* 115: 3086–3094, 2007.
86. Wojtaszewski JF, Mourtzakis M, Hillig T, Saltin B, Pilegaard H. Dissociation of AMPK activity and ACCβ phosphorylation in human muscle during prolonged exercise. *Biochem Biophys Res Commun* 298: 309–316, 2002.
87. Wolfe RR. The underappreciated role of muscle in health and disease. *Am J Clin Nutr* 84: 475–482, 2006.
88. Xu J, Li R, Workeneh B, Dong Y, Wang X, Hu Z. Transcription factor FoxO1, the dominant mediator of muscle wasting in chronic kidney disease, is inhibited by microRNA-486. *Kidney Int* 82: 401–411, 2012.
89. Yeo WK, McGee SL, Carey AL, Paton CD, Garnham AP, Hargreaves M, Hawley JA. Acute signalling responses to intense endurance training commenced with low or normal muscle glycogen. *Exp Physiol* 95: 35135–35138, 2010.
90. Zacharewicz E, Lamon S, Russell AP. MicroRNAs in skeletal muscle and their regulation with exercise, ageing, and disease. *Front Physiol* 4: 266, 2013.
91. Zacharewicz E, Della Gatta P, Reynolds J, Garnham A, Crowley T, Russell AP, Lamon S. Identification of microRNAs linked to regulators of muscle protein synthesis and regeneration in young and old skeletal muscle. *PLoS One* 9: e114009, 2014.